

To Arm or Not to Arm: an Arctic Dilemma

Lavinia Cavalet, UChicago XLab, 2024

Abstract:

The Arctic is not often under the geopolitical spotlight, due to its remote location, low levels of human activity and longstanding period of peace and cooperation following the fall of the Soviet Union. In most recent years, however, this forgotten region has witnessed a slow but steady increase in military investments, especially on behalf of the Russian Federation, one of five Arctic States. The motives behind states' renewed interest in the Arctic can be traced back to 2007, when a sudden exogenous shock caused a loss of ice greater than the size of Texas and California together. This shock unveiled a significant amount of natural resources – primarily oil and gas – previously hidden under kilometers of thick ice, and made them much easier to extract. Climate change and rising temperatures are making this once-hostile region much more accessible to Arctic and “near-Arctic” states, which are building the capabilities necessary to operate in this environment. With Russia doubling down on the modernization of old Soviet military bases, the region has become a new center for Great Power competition. While China has only recently become interested in accessing the Arctic's natural resources, Russia is much more familiar with this region's challenges and opportunities, and has engaged in power projection and hybrid threats to communicate its intentions to establish itself as a Great Arctic Power. The United States is lacking Arctic presence and operability after the dismantling of many military bases and infrastructure following the end of the Cold War. In this context, the United States is confronted with a new Arctic dilemma: how to respond to Russia's new threats in the High North? What is the optimal investment in Arctic military capabilities that will deter Russian threats without escalating tensions and increasing the risk of conflict? To answer these questions I use the framework developed by Charles Glaser in his famous paper, “When are Arms Races Dangerous? Rational vs Suboptimal Arming”. I evaluate the conditions required for three potential policies to succeed and then weigh the costs and benefits of each one. The three policies can be summarized as follows: 1. Arms Buildup Policy: the US chooses to build up its offensive capabilities in the Arctic, 2. Self-Restraint Policy: the US decides not to build up its military capabilities and maintain the current status quo, 3. ISR Investment Policy: the US opts to invest in domain awareness and surveillance capabilities rather than arms. After evaluating the ideal conditions for each policy to be the optimal policy, I reach the conclusion that the United States should invest in domain awareness and ISR capabilities in the Arctic to monitor Russia's activities and promptly react to potential military threats.

I. Introduction

Generally conceived as a bare land of ice and snow, inhabited only by polar bears, reindeers and ringed seals, the Arctic is not often associated with political instability and economic disputes, let alone warfare. Contrary to common knowledge, however, this region has barely been a theater of geopolitical peace. During the Cold War period, between 1947 and 1991, the Arctic was transformed into a key center for strategic competition between the United States and the Soviet Union, who deployed some of their most advanced weapon and surveillance systems near the Arctic region. Long range bombers and strategic submarines capable of launching intercontinental ballistic missiles (ICBMs), routinely patrolled the region, with the aim of collecting intelligence, projecting power and deterring the adversary. Up until the collapse of the Soviet Union, the Arctic was one of the most militarized regions in the world. With the end of the Cold War, important efforts were made to reduce the amount of both nuclear and conventional forces in the region, and several military bases were dismantled or closed down. This led many observers to conclude that the end of tensions between the US and the USSR would also end the strategic importance of the Arctic.¹ In 1996, the Arctic Council, the leading intergovernmental forum for cooperation in the Arctic was founded in Ottawa, Canada, and it was considered the ultimate sign that the region was finally entering a new, exceptional era of peace and regional stability - the era of *Arctic Exceptionalism*. Less than two decades since, this era was already over.

In 2007, after the discovery of vast amounts of oil and gas stored in the sea bed of the Arctic Ocean, Russia sent two mini submarines to the North Pole and planted a one meter-high titanium Russian flag on the underwater Lomonosov ridge, which Moscow claimed as a direct extension of its continental shelf. This symbolic move represented the beginning of the end of West-Russia cooperation in the Arctic, and was followed by a slow but steady increase in Arctic military spending by the Russian Federation. Old Arctic bases have been modernized, new ones are now under construction, and the Northern Fleet, one of 5 Russian military districts, has been repopulated with new, technologically advanced naval forces. Russian patrols near the borders of both Canada and Norway have been reported in greater numbers ever since, and Russia has bumped up its military training and exercises in the Arctic. This behavior was not well received by the rest of the Arctic states (US, Canada, Denmark and Norway), who have been increasingly worried about Russia's activities near their homeland. Such concerns were not, however, matched by an equally ambitious arming plan, and the West continued to invest little in Arctic capabilities, leaving a power asymmetry in the region. The Russo-Georgian war in 2008, as well as the annexation of Crimea in 2014 further strained Russia's relations with the US and its Western allies, but it wasn't until 2022, when Russia launched a full-scale invasion of Ukraine, that Arctic cooperation officially ended. Following the war in Ukraine, the Arctic Council's activities were halted, and Finland and Sweden requested to join the North Atlantic Treaty Organization (NATO), fearing that they would be Russia's next targets. This only enhanced Russia's fear of encirclement, and gave the country one more reason build-up arms in the Arctic.

¹Rob Huebert, "A New Cold War in the Arctic?! The Old One Never Ended!," 2019, https://arcticyearbook.com/images/yearbook/2019/Briefing-Notes/2_AY2019_COM_Huebert.pdf.

As the Russian Federation ramps up its military investment in the Far North, the Arctic security environment echoes memories and fears from the Cold War. Some researchers refer to the current state of affairs as a New Cold War in the Arctic, others argue that the old one never ended. Either way, the increasing tensions between the US and Russia, and the (re)militarization of the Arctic region, are becoming increasingly concerning for many analysts and officials, who are now facing some important and challenging questions: what is the optimal response to Russia's new threats in the High North? Should the US build up arms in the region? If so, how much? What kind of arms? Is arming at all necessary? In this paper, I aim to answer these questions, and more. As the United States considers the best course of action to respond to Russia's threats, it faces a fundamental dilemma: will arming lead to greater homeland security or will it initiate an arms race? I take a theory-based approach to solving the Arctic dilemma by determining whether an arms race in the Arctic is optimal or suboptimal for the United States. I will do so by applying Charles Glaser's strategic choice theory of arming to the Arctic geopolitical context. First, I discuss US and Russian interests and capabilities in the Arctic to determine how valuable the region is to the two states from an economic and military perspective. Then, I describe Glaser's arming model and explain how I intend to apply it to the unique Arctic security environment. From the model I extrapolate three potential policy options: 1. An Arms Build-up Policy, 2. A Self-Restraint Policy and 3. A Domain Awareness Policy. The first two are the direct outcomes of Glaser's framework, while the third one is my own contribution. After I evaluate the conditions under which each policy would be the optimal policy for the United States, I make a recommendation for which policy best applies to the current Arctic conditions and weigh the costs and benefits of pursuing such a policy.

II. Background

How has the Arctic come to be this new hotspot for economic and strategic competition? I identified three major causes to explain the recent surge in military activity and competition in the Arctic: natural resources, sea routes, scientific research and a new war in Europe. The first three causes are the direct result of climate change, which is causing the Arctic region to warm four times faster than the rest of the planet. Melting sea-ice is opening up new economic opportunities in the Far North and it is making this remote and harsh location much more easily accessible to human activity. Some forecasts predict that the Arctic ocean will be ice-free during the Summer by the next decade or so, allowing ships to transit much more easily. The latter cause is directly related to the Russian invasion of Ukraine in 2022. The war has had substantial spillover effects into the Arctic and sub-Arctic regions, starting from the cease of cooperation among Arctic Council states, to Russia's coercions through hybrid threats. I will now briefly explain each of these four major factors and how they contribute to increasing Arctic tensions.

Natural Resources

In 2007, the planet experienced one of the warmest heat waves in history. The rise in temperatures caused a rapid and unexpected decline in Arctic sea ice, exposing 1 million square miles of ocean, an area larger than Texas and California combined.² This exogenous

²Jonathan N Markowitz, *Perils of Plenty : Arctic Resource Competition and the Return of the Great Game* (New York, Ny: Oxford University Press, 2020), 49.

shock exposed vast amounts of natural resources until then hidden under thick ice. The next year, the US Geological Survey released a document arguing that 30% of the world's undiscovered gas reserves and 13% of the world's undiscovered oil are found in the Arctic.³ Nearly 80% of these resources are offshore in or beyond states' exclusive economic zones (EEZs), where states have to compete for ownership.⁴ States rushed to make claims over different parts of the Arctic's seabed; they started sending their best geological scientists to the North Pole to map the ridge and collect evidence that could demonstrate that those areas were part of their continental shelf, claims that would often overlap. Maybe the most contested area is the Lomonosov ridge, an underwater mountain range that spans 1,100 miles from the New Siberian Islands all the way to the Ellesmere Island of the Canadian Arctic Archipelago. Sovereignty over the ridge would give the sovereign state exclusive rights to fishing, shipping and extracting natural resources, according to the UN Convention on the Laws of the Sea (UNCLOS). While states have been using legal means to advance such claims for decades, some of them have also resorted to power projection to achieve their objectives and coerce other states.



Territorial Disputes in the Arctic⁵

In his book, *Perils of Plenty*, Jonathan Markowitz, professor of political science at USC, compared Arctic states' activities and investment in the region before and after the 2007 shock to determine how the newly available resources affected states' relations and competition. He finds that states economically dependent on natural resources – Russia and Norway – were

³ D. L. Gautier et al., "Assessment of Undiscovered Oil and Gas in the Arctic," *Science* 324, no. 5931 (May 28, 2009): 1175–79, <https://doi.org/10.1126/science.1169467>.

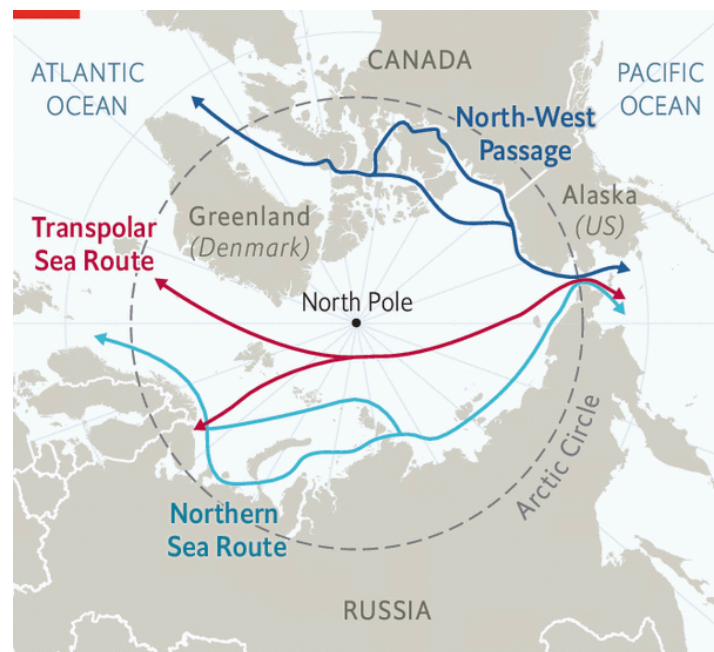
⁴ Jonathan N Markowitz, *Perils of Plenty : Arctic Resource Competition and the Return of the Great Game* (New York, Ny: Oxford University Press, 2020), 2-3.

⁵ The Economist, "Frozen Conflict," *The Economist* (The Economist, December 17, 2014), <https://www.economist.com/international/2014/12/17/frozen-conflict>.

more likely to project power in the Arctic after 2007 as a way of backing their economic interests in the region. Both states increased their Arctic military presence to improve their bargaining position vis-a-vis other Arctic states and to signal the strength of their interests in the region. Even China, which self-declared a “near Arctic state”, started to invest in Arctic-specific capabilities and infrastructure to make sure it doesn’t miss out on the region’s new economic opportunities. Hence, the discovery of vast reserves of natural resources strains tensions between Arctic states, who find themselves in a constant fight to control those resources, sometimes diplomatically and sometimes through demonstrations of power.

Sea Routes

Melting Arctic ice is also making waterways much more easily accessible, creating new economic opportunities. There are three main sea routes in the Arctic region: the Northern Sea Route, which runs along the Russian Arctic border, the North-West Passage, which skirts both Alaska and the Canadian archipelago, and the Transpolar Sea Route, which passes through the Bering Strait and cuts directly through the North pole.



Arctic Sea Routes⁶

These routes connect east Asia with Europe in far less time than any other major maritime route. Studies show that Arctic routes are 30% to 50% shorter than the Suez Canal and Panama Canal routes, with transit time reduced by 14 to 20 days.⁷ This, in turn, significantly reduces the costs associated with intercontinental shipping, especially those associated with

⁶ “How Viable Is Arctic Shipping?,” The Economist, January 18, 2024, <https://www.economist.com/the-economist-explains/2024/01/18/how-viable-is-arctic-shipping>.

⁷ Jill Kimball, “Melting Arctic Ice Could Transform International Shipping Routes, Study Finds,” Brown University, November 15, 2023, <https://www.brown.edu/news/2022-06-22/arctic#:~:text=According%20to%20Lynch%2C%20previous%20studies>.

fuel, and creates new profitable opportunities for many states and shipping companies. The Northern Sea Route (NSR) is the route that will likely become ice-free sooner – it is already ice-free during most of the Summer – and this has created disputes between the United States and Russia over the status of portions of the NSR: the US claims that they should be international waters, while Russia contends that it is within the Russian Exclusive Economic Zones (EEZ). The US has another similar dispute with Canada over the North-West Passage (NWP), with the US claiming that freedom of navigation applies to the NWP. The Transpolar Sea Route (TSR) remains a hypothetical route, as even with receding ice, the North Pole remains frozen throughout the year. However, as the Arctic becomes more navigable, states have greater incentives to compete over the legal status of these waterways; the legal battle has already begun, and prospects for an actual armed battle may be coming in the next decade or so.

Scientific Explorations

Scientific exploration has become a significant factor in the rising tensions in the Arctic. As climate change rapidly alters the Arctic's physical environment, it impacts not only the ice and ocean dynamics but also the biodiversity that thrives in this unique region. In response, states are ramping up scientific expeditions to closely monitor these environmental changes. These missions focus on tracking shifts in ice thickness, studying changes in species distribution, and understanding the broader ecological impacts of a warming Arctic. Moreover, scientific research is the foundation of all legal claims over the Arctic seabed and waterways. To support these claims, extensive scientific data is required, particularly concerning the boundaries of continental shelves. As a result, nations have significantly increased their scientific presence in the Arctic, conducting detailed surveys and mapping efforts to substantiate their territorial claims over areas that remain part of the international commons. However, a less known implication of increased scientific research is that many scientific research facilities in the Arctic are dual-purpose. While their primary mission is to gather environmental and geological data, these installations are often equipped with the capability to collect intelligence on other states' activities in the region. This dual-use nature of scientific expeditions and facilities adds a layer of complexity to Arctic geopolitics. What appears on the surface as purely scientific research can also serve strategic purposes, contributing to the broader contest for influence and control in the Arctic. The intertwining of scientific exploration with geopolitical and military objectives heightens tensions, as states are increasingly wary of each other's activities under the guise of research.

Ukraine-Russia War

The Russo-Ukrainian war that began in 2022 has had significant spillover effects in the Arctic. Following the outbreak of the war, Finland and Sweden requested to join NATO, seeking the protection and security that NATO offers its members. Although Russia has not yet responded militarily to NATO's enlargement, likely due to its ongoing preoccupations in Ukraine, in February 2024, Maria Zakharova, spokesperson for the Russian Foreign Ministry, announced that Russia would closely monitor Sweden's actions as part of the "aggressive military bloc" and would consider retaliatory measures of a military and technical nature. Years earlier, President Putin had issued a warning to Finland, stating, "When we look across the border now, we see a

Finn on the other side. If Finland joins NATO, we will see an enemy." While Russia's reaction has thus far been limited to rhetoric and threats rather than concrete actions, the Arctic region and the Nordic states have nonetheless experienced a noticeable increase in hybrid activities since the war began. These activities include cyberattacks, disinformation campaigns, and covert operations aimed at destabilizing the region and undermining the security and cohesion of NATO members. Russia's fear of NATO encirclement has therefore been an additional reason for Russia to step up its nuclear posturing in the High North. Moscow reportedly relocated several nuclear-capable bombers to the Kola Peninsula prior to last October's nuclear drills, which included tests of all three legs of the Russian nuclear triad in the Arctic.⁸ As Russia attempts to meet the apparent increased threat in Northern Europe while its army remains tied down elsewhere, it is likely to double down on aggressive nuclear signaling in the region.⁹ Russia will also seek to increase conventional deterrence along its northwestern flank, strengthening anti-access and aerial denial defenses around the Kola Peninsula.

The above factors all contribute to increasing tensions in the region. But how shall the US respond? The next section will first discuss what has been said so far on the US ideal response to new threats in the High North by scholars and government officials. Then, I will explain how I will seek to answer this question by using a famous theoretical framework developed by Charles Glaser to explain when arming is the optimal response and when it isn't.

III. Literature Review and Glaser's Framework explained

The literature on U.S. presence in the Arctic consistently underscores the importance of enhancing access and operational capabilities in the region. Scholars and policymakers alike argue that increased investment in Arctic infrastructure, including the establishment of new bases and modernization of existing facilities, is crucial. As emphasized by Schultz at the Wilson Center, "Presence equals influence. If we don't have a presence there, our competitors will," highlighting the strategic imperative of maintaining a robust U.S. presence to ensure successful deterrence.¹⁰ A recent report by RAND highlights the US lack of assets and infrastructure capable of operating in the Arctic, particularly in comparison with Russia. To better secure US interests in the Arctic, the report advocates for greater investments in platforms, infrastructure, equipment, communications, multi domain awareness, and enhanced tactics and training.¹¹ Among the most critical assets and infrastructure that different papers and reports recommend

⁸Thomas Nilsen, "Russia Steps up Military Posturing in the Arctic ahead of NATO's Nuclear Drill," The Independent Barents Observer, 2022, <https://thebarentsobserver.com/en/security/2022/10/russia-escalates-arctic-military-posure-ahead-natos-nuclear-drill>.

⁹Nicholas Lokker and Heli Hautala, "Russia Won't Sit Idly by after Finland and Sweden Join NATO," War on the Rocks, March 30, 2023, <https://warontherocks.com/2023/03/russia-wont-sit-idly-by-after-finland-and-sweden-join-nato/>.

¹⁰David Vergun, "Coast Guard Commandant Wants Bigger Arctic Presence – How Cool Is That?," U.S. Department of Defense, December 4, 2018, <https://www.defense.gov/News/News-Stories/Article/Article/1705544/coast-guard-commandant-wants-bigger-arctic-presence-how-cool-is-that/>.

¹¹Scott Savitz and Abbie Tingstad, "The U.S. Military Needs to Build Arctic Capabilities and Capacity," www.rand.org, December 21, 2023, <https://www.rand.org/pubs/commentary/2023/12/the-us-military-needs-to-build-arctic-capabilities.html>.

investing in are new icebreakers, US Coast Guard's Polar Security Cutters and a deep water port. These investments will be critical to ensure both military access as well as search and rescue operations. Failing to prioritize the region may result in multiple types of risks, including international perception of a security void, tacit tolerance of nefarious Russian and Chinese activities, loss of life, damage to national security, diminished economic potential, environmental degradation, and loss of resilience among resident communities.¹² Moreover, a recent policy brief by Brookings suggested that the US would require "significant investment in polar satellite coverage and other long-range communication facilities in Alaska and allied countries to sustain any future deployments and to ensure that the region remains attractive for private economic development that can benefit from similar communications infrastructure".¹³ The 2022 US National Strategy for the Arctic Region also recognizes the need to respond to the new challenges posed by climate change and increased activity in the region. To advance US interests, the strategy proposes four pillars spanning both domestic and international issues: 1 - Security, 2 - Climate Change and Environmental Protection, 3 - Sustainable Economic Development and 4 - International Cooperation and Governance.¹⁴ The highest priority in the document is therefore the need to deter threats to the US homeland and allies. To achieve this, the first pillar suggests improving domain awareness, expanding military presence through regular exercises and an increased icebreaker fleet, and strengthening cooperation with allies, particularly to counter Russian aggression. The strategy also emphasizes the importance of collaboration with local communities and resilience-building against climate change and cyber threats. In the more recent Arctic Strategy document from July 2024, these needs are reinforced. Specifically, the need to enhance domain awareness and Arctic capabilities, engage with Allies, partners and key stakeholders and exercise tailored presence in order to ensure successful integrated deterrence against Russia and the People's Republic of China.

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While researchers and analysts recognize the need for the US to respond to Russia's threats by increasing Arctic presence and operability, it isn't always clear what kind of infrastructure they should build, whether it should be predominantly defensive or offensive, and what are the costs and the benefits associated with different approaches. To bring some clarity to these arguments and provide a framework for determining what the best policy option is for the US, I will use Charles Glaser's theory elaborated in his influential paper, "When are Arms Races Dangerous? – Rational versus Suboptimal Arming".¹⁶ Here, Glaser develops the strategic choice theory of arming, aimed at providing a baseline for when arming is the rational/optimal response in a given international security environment. The two policy options resulting from this

¹²Abbie Tingstad et. al, "Report on the Arctic Capabilities of the US Armed Forces," *RAND*, 2023, 71.

¹³Jeremy Greenwood, "GREAT POWER COMPETITION and OVERSEAS BASING in the ARCTIC Executive Summary," *Brookings*, 2023, https://www.brookings.edu/wp-content/uploads/2023/02/FP_20230207_arctic_basing_greenwood.pdf.

¹⁴The White House, "National Strategy for the Arctic Region" (The White House, October 2022), <https://www.whitehouse.gov/wp-content/uploads/2022/10/National-Strategy-for-the-Arctic-Region.pdf>.

¹⁵US Department of Defense, "2024 Arctic Strategy," 2024, <https://media.defense.gov/2024/Jul/22/2003507411/-1/-1/0/DOD-ARCTIC-STRATEGY-2024.PDF>.

¹⁶ Charles L. Glaser, "When Are Arms Races Dangerous? Rational versus Suboptimal Arming," *International Security* 28, no. 4 (April 2004): 44–84, <https://doi.org/10.1162/0162288041588313>.

framework are: 1. An Arms Race Policy, whereby the optimal decision for the US is to build up its military capabilities in the Arctic, and 2. A Self Restraint Policy, which suggests that the US would be better off maintaining the status quo and not militarizing. Based on the distinctive features of the Arctic region and the yet not so high probability of conflict onset, I propose a third option which I call the Domain Awareness/ISR Policy. This policy entails investments in intelligence, surveillance and reconnaissance (ISR) capabilities in the Arctic, while maintaining low armament levels. ISR is defined as the coordinated acquisition, processing, and dissemination of accurate, relevant, and timely information and intelligence to support a commander's decision-making process.¹⁷ ISR systems can collect information from a wide range of sources, including electronic communications, optical, radar, and infrared imaging. Some of the assets used to collect this data include satellites, sensors, Unmanned aircraft, aviation systems, specialized ground, sea, or space-based equipment, and human intelligence teams.¹⁸

Now, I will summarize the “strategic choice theory of arming” model and explain how it applies to the Arctic case study. Glaser begins with one central question: is a security-seeking state more secure if both it and its adversary initiate arms buildups or if neither does? This game theory question is followed by three more specific and practical questions:

1. Would an arms race enhance the state's military capabilities—that is, its ability to perform military missions?
2. Would increased military capabilities decrease the adversary's security?
3. Would the benefits of the increased capabilities more than offset the dangers created by the adversary's insecurity?

To answer these questions and determine whether building up arms is a state's optimal policy decision, one must analyze material variables and information about the adversary's motives.

Material Variables

There are only two material variables in Glaser's model: power, and the offense-defense balance. According to Glaser, power is defined as the ratio of a state's resources that can be converted into military assets, and it determines whether a state is able to maintain a lead in a quantitative arms race. An arms build-up is often more advantageous or desirable when states are evenly matched in power or when one state has a significant power advantage. The offense-defense balance (ODB) is the ratio of the cost of forces required to take territory, to the cost of forces deployed by the defender. A state's prospects for acquiring an effective capability can be calculated by multiplying a quantitative variable of a state's power by the ODB. The greater the advantage for defense, the smaller the ratio of forces a state needs to achieve an adequate defensive capability.¹⁹

In this paper, I will approximate the power variable to be a country's defense budget adjusted for purchasing power parity (PPP). I also reconstruct the concept of the

¹⁷“What Is Intelligence, Surveillance, and Reconnaissance (ISR)? - CSMI,” CSMI Technology Services, August 2, 2022, <https://csmi.com/what-is-isr/>.

¹⁸Ibid

¹⁹ Charles L. Glaser, “When Are Arms Races Dangerous? Rational versus Suboptimal Arming,” *International Security* 28, no. 4 (April 2004): 52-53, <https://doi.org/10.1162/0162288041588313>.

offense-defense balance to better suit the unique context and geopolitical environment of the Arctic region. The Arctic is not a conventional war theater; weather conditions are very extreme, and there is no real land to operate on, all the hard surface is thick ice and the rest is the Arctic ocean. For these reasons, it is more difficult to operationalize the ODB, which better applies to land warfare. If a war was to break out in the Arctic, it would be a war predominantly fought over sea and air, so having quantitatively and qualitatively superior maritime and air forces gives states an incredible advantage. Sea – and to a lesser extent air – forces are also difficult to classify as either offensive or defensive. As explained by the Salvo model of missile combat for modern naval operations, whoever “attacks effectively first” wins.²⁰ This suggests that, in a war where missiles are the main weapons used to inflict unacceptable damage to the adversary, it doesn’t matter whether a state’s capabilities are offensive or defensive in nature, as long as they are able to accurately strike the adversary’s targets. This is true because both sea control and sea denial operations are dominated by precision guided munitions. Hence, for the purpose of this paper, I will assume that there is no real distinction between offensive and defensive capabilities in the Arctic. In such a war, ISR becomes extremely important. While ISR systems are commonly associated with defensive capabilities, as many of them are in fact used in missile defense, they also become critical assets in an offensive operation, as they allow states to quickly and accurately target and take down an adversary’s strategic infrastructure.

Given the unique characteristics of the Arctic, quantitatively measuring the offense-defense balance (ODB) becomes challenging. To simplify the analysis, I will define the ODB as the relative ease, effectiveness, and cost of launching an offensive operation versus defending against an attack. Additionally, I will treat offensive and defensive capabilities as largely indistinguishable due to the nature of missile combat, where the advantage is given by the ability to effectively strike the opponent’s targets, reducing the importance that capabilities are offensive or defensive in nature. Thus, whether the ODB favors offense or defense will depend primarily on each state’s interests and capabilities in the region, reflecting their willingness to overcome the technical and logistical challenges of Arctic operations. To measure the ODB, I will compare the naval, air, and ISR (Intelligence, Surveillance, and Reconnaissance) capabilities of the US and Russia, focusing on both the quantity and quality of assets. This comparison will help determine which state holds a strategic advantage, allowing me to conclude whether the ODB favors defense or offense for the US in the Arctic.

Information about the adversary’s motives

A state’s information about its adversary’s motives also influences the state’s decision to arm. Glaser describes this variable as the probability that the adversary is a security seeker and not a greedy state. This definition is strictly related to the security dilemma, a concept used in international relations to explain how states’ arming decisions might lead to war. More specifically, it explains how the decision by one state to build up arms for security reasons can lead other states to fear for their own security, causing them to pursue arming policies that

²⁰ The Salvo model of naval missile combat is a quantitative framework that simulates engagements between opposing naval forces by modeling the exchange of missile salvos. It specifically assesses the probability of ships being hit or destroyed based on factors like the number of offensive missiles launched, the effectiveness of defensive countermeasures (e.g., anti-missile systems or electronic warfare), and the damage capacity of the ships.

increase tensions and potentially escalate into conflict. A security-seeking state is therefore a state that seeks to protect its territory from other states that are perceived as threatening their homeland security or key strategic interests. A state that faces a security dilemma must therefore balance between two options: 1) building up arms to provide greater military capability and communicate resolve, but risk signaling malign motives, and 2) restraint from building arms and use cooperative means of bargaining that make the state less capable to respond militarily to threats, but signal benign motives. If a state believes its adversary is security-seeking, the danger created by a shortfall in military capabilities is smaller, and the benefits of communicating benign motives are larger. When states are quite sure that others are security seeking, information about motives can essentially eliminate the security dilemma, even when material conditions would suggest the opposite.²¹ On the other hand, if a state believes that its adversary is primarily motivated by greed, then building up arms and acquiring offensive capabilities becomes the more attractive strategy. It is therefore the combination of information and material variables that determines a state's preferred arming policy.

Offense-Defense Balance		
	Defense > Offense	Offense > Defense
Advantage	• No need to build—defensive doctrine without force advantage is adequate; action-reaction peters out • Exceptions if geopolitical rationales generate offensive requirements ♦ Must then compare political costs against military benefits (depends on information about adversary's motives)	• Arms buildups and offensive doctrine often best, unless • Power advantage is large enough to make defense possible • Power advantage is too small to guarantee success; weigh risks of losing a race • Offense and defense are different; weigh the benefits of limiting offense against risks of cheating
Power		
Disadvantage	• If defense advantage is greater than power disadvantage, defensive doctrine is still feasible; arms buildups can be avoided or should peter out • If defense advantage is less than power disadvantage, cannot prevent adversary from acquiring offensive capability ♦ Weigh arms competition to retain some deterrent capability against cooperating to improve relations	• Buildup holds little prospect of producing a defensive capability • Might build up to preserve an offensive capability

Material Variable and the Choice of Arming

Glaser's framework acknowledges the implications provided by both the deterrence and spiral models, but it also emphasizes the impact of uncertainty about the adversary's motives. Consequently, the theory advocates for a mix of spiral and deterrence model policies, except in

²¹ Charles L. Glaser, "When Are Arms Races Dangerous? Rational versus Suboptimal Arming," *International Security* 28, no. 4 (April 2004): 56-57, <https://doi.org/10.1162/0162288041588313>.

cases when the adversary's intentions are clear. In this paper, I will do my best to evaluate both material variables and state's motives of the US and Russia when it comes to the Arctic, with the aim of determining the US optimal policy option to respond to Russia's militarization of the region. I will explore such policies in more detail in section V, where I investigate the conditions under which each policy is the optimal policy and draw my conclusions for what policy better suits the Arctic geopolitical conditions and the US interests. Before I do that, I must address the Arctic states' interests and capabilities in the region. The next section will therefore discuss the US, Russia and other Arctic states' economic and strategic interests, and report on each state's military capabilities to operate in the Arctic and protect their stakes in the region.

IV. Interests and Capabilities of Arctic states

In order to determine how to respond to Russia's military expansion in the Arctic region, it is crucial to evaluate the interests and ambitions of the two Arctic states. The need to do so is driven by the importance of understanding the willingness of a state to resort to violence to protect its claims in the region; the higher the stakes a state has in the Arctic, the greater we can expect their willingness to use force to protect those stakes. In other words, if a state perceives the Arctic to be a vital location for the achievement of its economic, national or strategic objectives, its motivation to undertake an aggression against another state actor that interferes with the achievement of those goals will be greater. The analysis of states' interests needs to be paired with an evaluation of states' capabilities to operate in the Arctic environment and to defend those interests. A comprehensive assessment requires evaluating Arctic military capabilities across different domains: mainly maritime, air, and ISR systems. In this section I will therefore first discuss the US and Russia's current interests and objectives in the region, and then evaluate each state's ability to operate in such a hostile region.

4.1: Arctic states' interests

A. Russia: The Arctic State *Par Excellence*

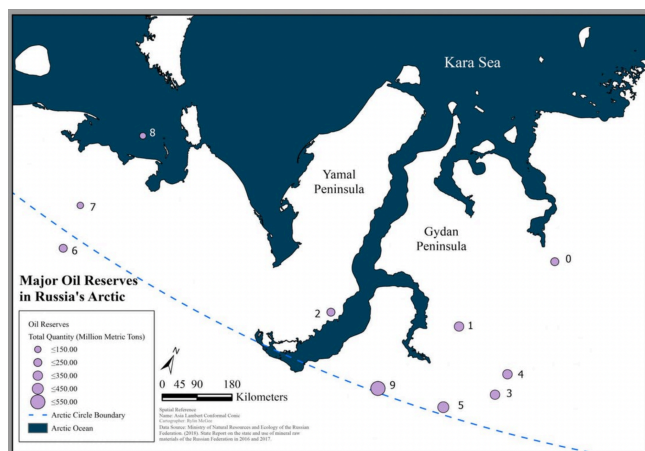
Claiming 53% of the Arctic Ocean's coastline, the Russian Federation considers the Arctic to be an integral part of its national identity. From the early explorations by Pomor settlers along the northern coastlines, to Soviet-era ambitions of Arctic dominance, the region has been seen as a symbol of national pride and strategic importance. Moreover, its vast natural resources and strategic sea routes have further cemented its role in Russia's economic and geopolitical ambitions. Approximately 2.5 million people live in the Russian Arctic, accounting for almost half of the population living in the Arctic worldwide.²² Therefore, the economic development of this region is an absolute priority for the Russian Federation, which raised the

²²"The Russian Federation," Arctic Council, 2023, <https://arctic-council.org/about/states/russian-federation/#:~:text=The%20extraction%20of%20natural%20resources%2C%20primarily%20oil.>

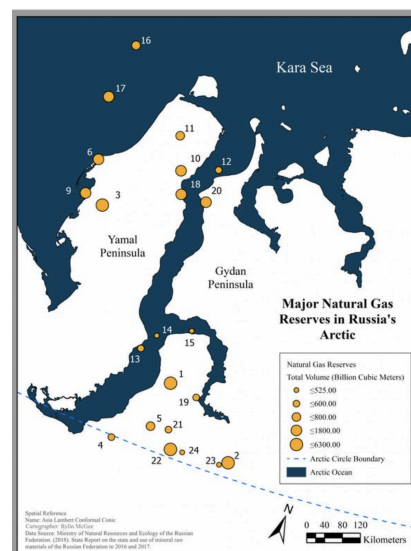
Arctic to its second priority region after Russia's "near abroad" in the 2023 Foreign Policy Concept.²³

Economic interests

From an economic point of view, the Arctic constitutes nearly 20% of Russia's GDP, 22% of its exports and more than 10% of all investment in Russia.²⁴ The Arctic economic revenues come from three main sectors: energy resources extraction, maritime transportation and food security, with the energy sector being by far the leading source of Arctic regional development. With 60% to 70% of Russia's export earnings deriving from the extraction, production and export of energy, Russia is undoubtedly the most land-oriented Arctic state.²⁵ The Russian Arctic alone comprises over 35,700 billion cubic meters of natural gas and over 2,300 million metric tons of oil and condensate with the majority concentrated in the Yamal and Gydan peninsulas.²⁶ While Russia's main oil and gas reserves lie on or near the Siberian coast, the US Geological Survey estimates that 30% of the world's gas reserves and 13% of total undiscovered oil are found in the Arctic, 80% of which are offshore, in or beyond states' exclusive economic zones (EEZ). This makes room for competition over territories still classified as international waters, with states trying to demonstrate that the North Pole is an extension of their own continental shelf.



Map of Russia's major oil reserves in the Arctic²⁷



Map of Russia's major gas reserves in the Arctic²⁸

²³Russian Federation, "The Concept of the Foreign Policy of the Russian Federation - the Ministry of Foreign Affairs of the Russian Federation," mid.ru, March 31, 2023, https://mid.ru/en/foreign_policy/fundamental_documents/1860586/.

²⁴ Ian Williams et al., "America's Arctic Moment: Great Power Competition in the Arctic to 2050," Csis.org, 2020, <https://www.csis.org/analysis/americas-arctic-moment-great-power-competition-arctic-2050>.

²⁵Jonathan N Markowitz, *Perils of Plenty : Arctic Resource Competition and the Return of the Great Game* (New York, Ny: Oxford University Press, 2020), 87.

²⁶ Rylin McGee, "Mapping Russia's Arctic Hydrocarbon Development Scheme," The Arctic Institute, February 18, 2020, <https://www.thearcticinstitute.org/mapping-russia-arctic-hydrocarbon-development-scheme/>.

²⁷ Ibid

²⁸ Ibid

The large and consistent revenues generated by land extraction, paired with failed efforts to diversify its economy have caused Russia to double down on the extraction of oil and natural resources, becoming “addicted” to land rents. According to the Rent-Addiction Theory developed by Jonathan Markowitz, land-oriented states like Russia are more likely to pursue exclusionary foreign policies and project power to ensure access to resource-rich territories. This theory is important in the analysis of Russia’s interests in the Arctic, as it points to the country’s willingness to use military force to protect its economic stakes in the region. When the 2007 climate shock untapped a new breadth of natural resources, until then hidden under kilometers of thick Arctic ice, the Arctic region acquired even more value for Russia, becoming a key location for the future economic growth of the country. Arctic development could potentially increase Russia’s known oil deposit by another half and could account for more than 20% of Russia’s total oil output by 2030.²⁹ The economic benefits of investing in the Arctic are therefore extremely critical for the preservation of Russia’s great power status. This is even more so today, as the rising temperatures and increasing ice melting rates make this vast region more accessible and subject to competition among not only Arctic states but also “near Arctic states” like China. In light of increased competition over natural resources, Russia feels the need to scale up its presence and activity in the region and ensure that it doesn’t lose the comparative advantage it gained over decades of prioritizing the Arctic.

While the extraction of energy resources is probably the leading driver of Russian economic interests in the Arctic region, exclusive access to the Northern Sea Route (NSR) is another strategic objective of the RF. The NSR extends for 13,000km from the Barents Sea, near the Russian border with Norway, to the Bering Strait between Siberia and Alaska, and it is the shortest shipping route connecting Western Eurasia and the Asia-Pacific region. The route is currently only viable during the Summer months, when the ocean along the Russian coastline is ice-free, and cargo ships and bulk carriers can more easily navigate the hostile Arctic waters. However, as climate change causes ice to thin and recede at a much faster rate, shipping traffic along the NSR has increased significantly. In the ten years between 2011 and 2021, the total cargo volume transported along the route increased from just above 3,000 to well over 34,000 mln t.³⁰ The NSR represents a substantial shortcut for the transportation of goods between Europe and Asia, with distance savings up to 50% higher compared to the currently used shipping lanes via Suez or Panama.³¹ Nonetheless, researchers and analysts are still very skeptical that the NSR will replace the most popular commercial lanes any time soon, due to the logistical challenges posed by the harsh Arctic conditions that require specific operational capabilities. They predict that only 5% of global shipping may be diverted to the Arctic by 2050.

Russia, who routinely uses the NSR for domestic and international shipping of petroleum, gas and mineral resources, considers the NSR to be under Russian sovereignty, as a portion of its waters lies within Russia’s exclusive economic zone. Under Russian law, the NSR is referred to as a “historically emerged, national transportation route of the Russian

²⁹ “Arctic Russia - Vostok Oil,” arctic-russia.ru, n.d., <https://arctic-russia.ru/en/project/world-class-cluster/>.

³⁰ “Shipping Traffic at the NSR in 2022 | Northern Sea Route Information Office,” June 9, 2023, <https://arctic-llo.com/nsr-2022-short-report/>.

³¹ Malte Humpert, “The Future of the Northern Sea Route - a ‘Golden Waterway’ or a Niche Trade Route,” The Arctic Institute, September 15, 2011, <https://www.thearcticinstitute.org/future-northern-sea-route-golden-waterway-niche/>.

Federation”, subject to a set of rules regulating shipping in the area.³² In an amendment to article 14.3 to the Law on Internal Waters made in 2022, the Russian government advanced claims that the NSR waters are not international waters, hence they are not subject to the UN Convention on the Laws of the Sea regulations on international navigation. Invoking Article 234 of UNCLOS - which states that coastal countries have the right to enforce laws and regulations in ice-covered areas within the limits of EEZ - Russia requires foreign warships and other government vessels to seek permission to enter the NSR zone at least 90 days in advance. Moreover, it mandates that no more than one foreign sovereign immune vessel may be in Russia’s NSR waters simultaneously, and that all foreign submarines sail on the surface under their flag.³³ Under such clauses the Russian government reserves the right to suspend the transit of warships if they perceive them as threats to Russia’s security.



The NSR in the context of Russia’s Arctic maritime area³⁴

The controversy surrounding Russia’s amendment is related to the legal status of the NSR straits. Russia claims that the four major straits of the Northeast passage - Kara Gate, Vilkitsky, Sannikov, and Laptev - to be internal waters and therefore subject to the Act’s requirements.³⁵ The United States, on the other hand, holds that they are international waters and that the regime of transit passage as articulated by UNCLOS applies to them. The adoption of the Act by the Russian government gives the US more reasons to conduct a Freedom of Navigation Operation in the NSR and transit along the route without asking for permission. Such an operation, however, entails significant geopolitical risks, as Russia may interpret it as an

³² Andrey Todorov, “New Russian Law on Northern Sea Route Navigation: Gathering Arctic Storm or Tempest in a Teapot?,” Belfer Center for Science and International Affairs, March 9, 2023, <https://www.belfercenter.org/publication/new-russian-law-northern-sea-route-navigation-gathering-arctic-storm-or-tempest-teapot>.

³³ Cornell Overfield, “Wrangling Warships: Russia’s Proposed Law on Northern Sea Route Navigation,” Default, 2022, <https://www.lawfaremedia.org/article/wrangling-warships-russias-proposed-law-northern-sea-route-navigation>.

³⁴ Ibid

³⁵ Ibid

attack to its homeland security and economic objectives. In deciding how to respond to Russia's claims in the Arctic, the US needs to consider the Russian economic stakes in the Arctic, and what they might be willing to do to protect them.

Not only is Russia motivated to pursue the economic development of the Arctic, but it is willing to project military power to protect its claims in the region and fight any world power that will try to challenge Moscow's territorial claims - both metaphorically and literally. The country's military build-up in the region is often associated with Russia's efforts to communicate ambitions, power, determination, and competence in order to strengthen deterrence and influence potential adversaries.³⁶ How far will Russia go to do so is unclear, but given the vital importance of this region for the Federation, the US will need to carefully evaluate how to contain Russian military expansion and prepare to respond to a potential Russian attack.

Military and Strategic Objectives

The Arctic is not only vital for Russia's future economic development, but it represents one of the most crucial locations for its homeland security and defense. The Russian Northern Fleet - one of five components of the Russian Navy - is headquartered in Severomorsk on the Kola peninsula, near the border with Norway and Finland. First established in 1933, the Northern Fleet's mission is to not only protect Russia's northwestern Arctic region, but most importantly ensure the survivability of its nuclear second-strike capabilities. The fleet accounts for most of the country's atomic ballistic missile submarines as well as major combatants and naval aviation units. The Northern Fleet also plays a key role in protecting vital Arctic infrastructure, including oil and gas installations, which are essential to Russia's economic ambitions in the region. As the Arctic continues to thaw and resource extraction becomes more feasible, this military presence ensures that Russia can protect both its strategic and economic interests.

Russia's strategic interests in the Arctic are largely defensive: defend its nuclear second-strike and deterrent capabilities, defend the Russian borders, and protect its economic interests, discussed in the previous section.³⁷ Russia's Arctic strategy for 2035, adopted in 2020, emphasizes three key objectives: 1. To ensure Russia's sovereignty and territorial integrity, 2. To develop the Russian Arctic as a strategic resource base and its rational use to accelerate national economic growth and 3. To develop the NSR as a competitive national transport line of the Russian Federation in the global market.³⁸ Russia officially states that it wants to achieve these objectives within established international laws, but it also justifies the expansion of its military presence in the region by highlighting the potential threat of debarment by foreign

³⁶ Katarzyna Zysk, "Russia's Military Build-up in the Arctic: To What End?," September 2020, 3.

³⁷ Colin Wall and Njord Wegge, "The Russian Arctic Threat: Consequences of the Ukraine War," *Www.csis.org*, January 25, 2023, <https://www.csis.org/analysis/russian-arctic-threat-consequences-ukraine-war>.

³⁸ Julian Meade, "Russia's New Arctic Policy 2035: Implications for Great Power Tension over the Northern Sea Route," 2020, https://ni-u.edu/wp-content/uploads/2023/11/NIUShort_07212020_DNI202201735_IceRusha.pdf.

governments or international associations of Russia's creation of legal businesses or other activities in the Arctic.³⁹

While Russia's ambitions are predominantly defensive, its aggressive rhetoric and far reaching territorial claims suggest that Russia also has offensive goals. Some analysts observe that Russia seeks to use the Arctic as a channel for power projection, especially into the North Atlantic Ocean through the GIUK Gap, which would pose an increased threat to NATO's sea lines of communication and carrier battle groups. The steady increase in military exercises and weapons testing in the Arctic is often seen as signaling Russia's arsenal superiority and operational readiness to deter and defeat adversaries in a potential conflict. Russia also employs hybrid tactics and threats to disrupt states' activities and increase the risk of escalation as a tool for coercive leverage, while carrying out actions that remain below the threshold of a military response. Since Russia's invasion of Ukraine in 2022, cyber operations have been on the rise, with states reporting several ransomware attacks on electoral systems across the Arctic region – notably in Norway, Alaska and Finland –as well as a considerable increase in hacking attempts into governmental systems.⁴⁰ The damage to the Nord Stream gas pipelines in September 2022 is also believed to be linked with a Russian attack, but the lack of evidence makes it difficult to confirm the attacker's identity. Russian espionage activities have also been increasingly more concerning, and Western countries have responded by expelling a large number of Russian intelligence officers – both Denmark and Norway sent back to Moscow 15 intelligence officers for engaging in espionage activities.⁴¹ Among the hybrid threats, Russia uses electronic warfare. Russia reportedly maintains electronic or electromagnetic warfare centers at Severomorsk, Kamchatka, and Primorsky Krai. These systems can interfere with key maritime communications systems, jam Western communications, radars and datalinks to blind the adversary and facilitate anti-submarine warfare. This capability can provide a battlefield advantage and be used maliciously both in peace and war time.⁴²

Russia's current militarization of the northern region mainly consists of defensive capabilities designed to protect its borders and second-strike assets through the so-called bastion defense system. Much of the growth in infrastructure is intended to conduct search and rescue operations and protect maritime shipping routes and economic investments projects.⁴³ However, Russia's military capabilities have an inherent offensive potential, and can quickly be redirected to threaten NATO allies if needed. Russia has increased investment in dual-use capabilities such as radar surveillance and drone bases, which are used for communication and security purposes but can easily be repurposed for offensive goals. For this reason, Russia's

³⁹ Scott Saviz et. al, "Report on the Arctic Capabilities of the US Armed Forces Appendixes," *RAND*, 2023, 4.

⁴⁰ Katarina Kertysova and Gabriella Gricius, "Countering Russia's Hybrid Threats in the Arctic," 2023, https://www.europeanleadershipnetwork.org/wp-content/uploads/2023/12/23_11_22_Countering-Russias-Hybrid-Threats-in-the-Arctic15_ES_EK40.pdf.

⁴¹ Ibid

⁴² Colin Wall and Njord Wegge, "The Russian Arctic Threat: Consequences of the Ukraine War," *Www.csis.org*, January 25, 2023, <https://www.csis.org/analysis/russian-arctic-threat-consequences-ukraine-war>.

⁴³ Eugene Rumer, Richard Sokolsky, and Paul Stronski, "Russia in the Arctic—a Critical Examination," *carnegieendowment.org*, March 2021, <https://carnegieendowment.org/research/2021/03/russia-in-the-arctica-critical-examination?lang=en>.

offense-defense balance is very fragile – the current defense-oriented strategy might easily shift towards an offensive posture, should the US and its allies boost up their missile defense systems and anti-submarine warfare (ASW) assets.⁴⁴

B. United States:

The United States officially became an Arctic state in 1867, when the US government finally purchased Alaska from the Russian Empire for over 7 million dollars. This purchase, considered very controversial at the time, represented the end of Russia's presence in North America and granted the US access to the pacific northern rim. Criticism was finally appeased in 1896 when major gold fields were discovered in the Klondike's region. During WW2 and the Cold War, Alaska acquired strategic value due to its vicinity to the North pole, which made the state critical to US national security. In a speech to Congress in 1935, General Billy Mitchell stated that "Alaska is the most strategic place in the world", referring to the short distances between Alaska and key strategic locations in Europe and Asia. This makes it the perfect platform for power projection into distant geopolitical theaters, and a crucial geographical location for early warning and missile defense systems. Today, Alaska accounts for only 0.2% of US total GDP, with Government, transportation and mining being the state's primary industries.⁴⁵

Economic interests

The Arctic has been of little economic interest to the United States in the past, and economic development is mentioned in the 2022 Arctic Strategy only in the context of improving the livelihoods of native Alaskan communities and supporting them through the energy transition. There is no mention of US interests and claims over territories and natural resources. This is consistent with Markowitz' Rent-Addiction Theory. As a democratic, production oriented state, the US is more interested in securing goods for its citizens by maintaining open access to markets globally rather than appropriating resources as a source of rents and revenue.⁴⁶ While Russia and even Norway heavily rely on resource extraction for their economic prosperity, the United States has been able to diversify its economy, and consistently diminished its dependence on land rents. In absolute terms, the US is considered a major energy producer, but relatively speaking, natural resource rents only account for about 1% of total American GDP as of 2021.⁴⁷ Contrarily, services made up almost 80% of the total US economy in the same year, highlighting the overwhelming dominance of the production-oriented sector. Moreover, extracting resources in the Arctic is extremely expensive and requires Arctic specific capabilities that the US is lacking, such as icebreakers. The US decision not to ratify the UN Convention on

⁴⁴ Eugene Rumer, Richard Sokolsky, and Paul Stronski, "Russia in the Arctic—a Critical Examination," carnegieendowment.org, March 2021,

<https://carnegieendowment.org/research/2021/03/russia-in-the-arctica-critical-examination?lang=en>.

⁴⁵ USAFacts, "Economy of Alaska Statistics and Data Trends: GDP Ranking, Unemployment Rate, and Economic Growth," USAFacts (USAFacts, August 30, 2023),

<https://usafacts.org/topics/economy/state/alaska/#0deb1089-9b10-4756-be6c-fe33dd9730b0>.

⁴⁶ Jonathan N Markowitz, *Perils of Plenty : Arctic Resource Competition and the Return of the Great Game* (New York, Ny: Oxford University Press, 2020), 125.

⁴⁷ "Total Natural Resource Rents," Our World in Data, 2024,

<https://ourworldindata.org/grapher/natural-resource-rents?country=~USA>.

the Law of the Sea (UNCLOS), also limits its ability to make extended continental shelf claims and thus exercise sovereignty over offshore reserves of natural resources.

Of slightly more importance for US economic interests are disputes over maritime routes accessibility. One of US strategic objectives in the 2022 Arctic Strategy is to protect the freedom of navigation across the Arctic and defend the international rule of law, especially in light of disagreements over the legal status of Arctic waterways. The US has two disputes in the Arctic: one with Moscow over a portion of the NSR, and a second one with Canada over the Northwest passage. In the US-Canada dispute, Canada claims the Northwest Passage as internal territorial waters, on the basis of a long history of Inuit use of the waters. The US, on the other hand, contends that it is a strait used for international navigation and therefore freedom of navigation applies. Similarly, Russia claims the NSR to be under Russian sovereignty, while the US argues that the NSR contains straits used for international navigation and therefore the regime of transit passage applies to passage through those straits. As ice melts and these maritime routes become increasingly more attractive for commercial shipping and transportation, it will become crucial for the US to resolve these disputes. Ratifying the UNCLOS could also help the US strengthen its arguments for freedom of navigation and overflight through the NSR and NWP.

Military and Strategic Objectives

While the US economic interests in the Arctic are relatively low, its strategic and military objectives in the region remain highly significant. As I mentioned earlier, the Arctic represents the shortest route between Alaska and both Asia and Europe, as well as the location where US borders are the closest to Russian territory. If Russia ever decided to strike the United States, it would do so over the Arctic. During the Cold War, the Arctic acted as a strategic arena for US-USSR competition and power projection, with both countries using the Arctic as a testing ground for their newest military technologies and nuclear missiles. For the United States, the region also served as their eyes and ears into Russia's activities, often using submarines to monitor Russia's activities and intercept communications. After the end of the Cold War, however, the US and Russia entered a new era of cooperation in the Arctic, the era of Arctic Exceptionalism, marked by peaceful relations and disarmament processes. Today, the Arctic remains a highly strategic place for the United States, and with Russia building up its military forces and nuclear capabilities, the importance of Alaska for missile defense and early warning systems will only increase.

The US security concerns are outlined in the most recent Arctic Strategy, released in July 2024. The document aligns with the interests and concerns discussed in previous national Security Strategies, highlighting climate change and the shifts in the geostrategic environment as the main drivers for a new strategic approach to the region. The United States seeks an Arctic region that is peaceful, stable, prosperous and cooperative, in which the US homeland remains secure and vital national interests are safeguarded.⁴⁸ The Strategy is aimed at strengthening the US ability to build integrated deterrence and manage risk in the region through 3 main channels: 1) enhancing domain awareness capabilities, 2) engaging with allies

⁴⁸ US Department of Defense, "2024 Arctic Strategy," 2024, <https://media.defense.gov/2024/Jul/22/2003507411/-1/-1/0/DOD-ARCTIC-STRATEGY-2024.PDF>.

and partners, and 3) exercising tailored presence. In light of US military and strategic interests in the region, the absence of readily available military capabilities deployed in the Arctic gives Russia a strategic advantage and reduces the US ability to deter and quickly respond to Russia's threats.

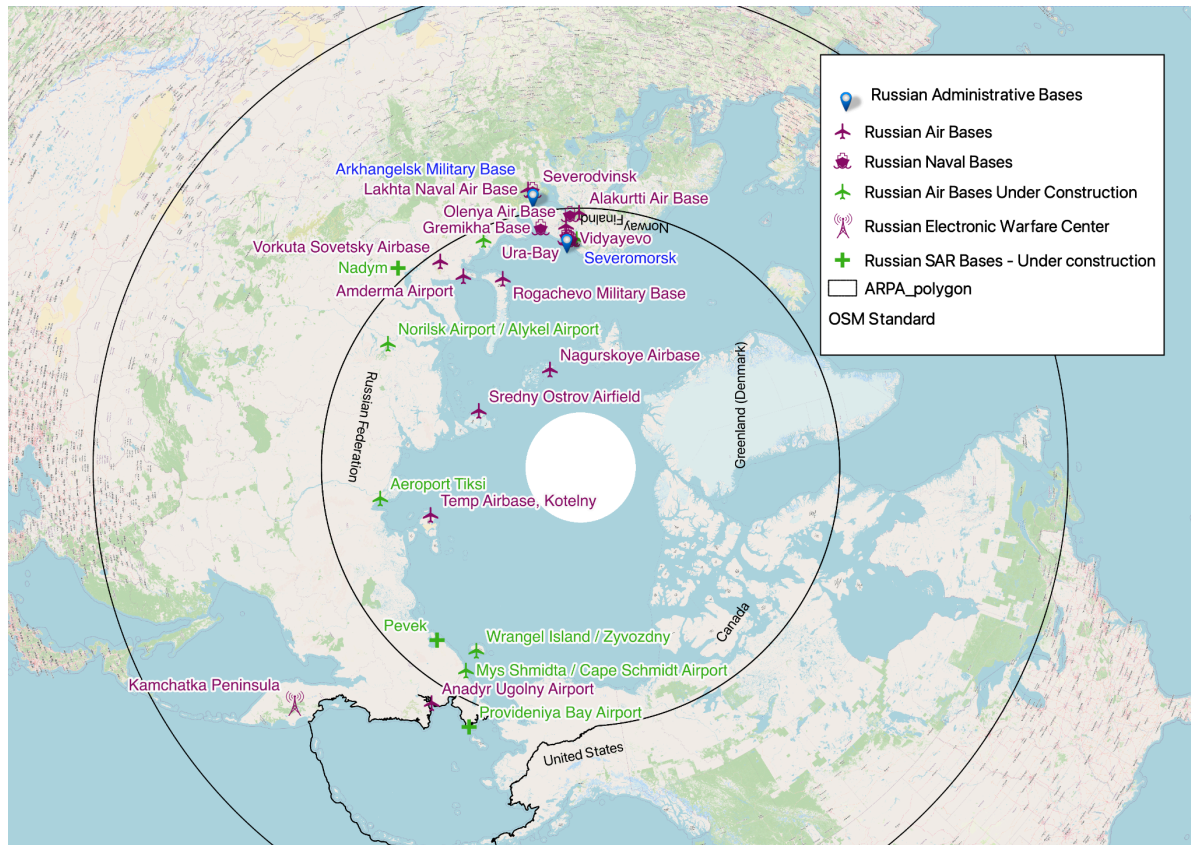
4.2: Arctic states' capabilities

When states have disputed or conflicting geopolitical interests in a particular region, they must back these interests with substantial military and operational capabilities. This is especially crucial in the Arctic, where extreme conditions demand specialized infrastructure, equipment, and highly trained personnel to ensure effective operations. In this section, I will compare the Arctic capabilities of the Russian Federation and the United States. The goal is to provide a comprehensive overview of each state's ability to operate in the Arctic across key domains, including military bases, naval, and air forces, missile defense systems, logistical support networks, and intelligence, surveillance, and reconnaissance (ISR) capabilities. By evaluating these areas, we can assess which state is better positioned to project power and secure its interests in the region.

4.2.1: Arctic bases

The Russian Federation has by far the largest number of Arctic military bases, with over twenty active bases and nine under construction or modernization. The vast majority of these bases are concentrated in the Western Arctic, near the border with Norway and Finland, where the Northern Fleet as well as Russia's nuclear second-strike capabilities are stationed. In the past decade, Russia has been refurbishing and modernizing its Soviet-era assets and constructing new facilities, including search and rescue (SAR) bases, airfields and radar stations. Throughout the AZRF Russia has reopened or expanded 13 airfields, ten radar stations, 20 border stations and ten SAR stations. Additionally, the volume of modern communication equipment for armed forces suitable for operation under Arctic conditions grew by 20% between 2012 and 2017.⁴⁹ These developments underscore Russia's commitment to maintaining and expanding its military influence in the Arctic, which is central to its broader strategic and defense objectives.

⁴⁹ Scott Saviz et. al, "Report on the Arctic Capabilities of the US Armed Forces Appendixes," *RAND*, 2023, 18.



Map of Russian military bases above the Arctic circle

The United States, on the other hand, only has one base above the Arctic circle, the Pituffik Space base, located in North-western Greenland. The remaining US bases are located in Alaska, with the most important ones being: Eielson Air Force Base, Clear Air Force Station, Fort Richardson, Fort Greely, Fort Wainwright, and USCG Juneau, home to the US Coast Guard. These bases play essential roles in the U.S. defense strategy, offering capabilities for air and missile defense, early warning systems, and rapid deployment of forces in the Arctic region. Their main purpose is that of monitoring and detecting potential threats before they become real dangers to the US national security, but they often lack the capacity to fully support sustained operations in the extreme Arctic environment. The limited number of bases, combined with the challenging conditions, means that the U.S. has less capability for year-round presence and rapid response compared to Russia's extensive network of Arctic bases. Additionally, the existing infrastructure in Alaska is primarily geared towards early warning and air defense, with less emphasis on the broader range of military operations that might be necessary in a rapidly changing Arctic environment.

4.2.2: Maritime Capabilities

The Arctic region is a maritime region, meaning that sea forces are crucial for operability. In this section I will compare US-Russia maritime capabilities including submarines, surface vessels and icebreakers. For Russia, such capabilities are predominantly located in the

Northern Fleet, which operates not only in the western most part of the Arctic, but also takes operational control over vast portions of the central and eastern Arctic. Russia's Pacific Fleet also covers the eastern-most part of the Arctic region.

The Northern Fleet's assets include a range of advanced submarines, such as the Borei-class equipped with intercontinental ballistic missiles and the Yasen-class multipurpose submarines, designed for both anti-ship and land-attack missions. These submarines provide Russia with significant strategic and tactical advantages, particularly in terms of nuclear deterrence and maritime dominance. In addition to its powerful submarine fleet, Russia's surface vessels in the Arctic include the Admiral Kuznetsov aircraft carrier, Slava-class cruisers, and Gorshkov-class frigates, all of which are equipped with a variety of anti-ship, anti-aircraft, and anti-submarine weapons. These ships are crucial for maintaining a strong maritime presence in the Arctic, enabling Russia to project power across the region. Russia's commitment to maintaining and expanding its maritime capabilities in the Arctic is further underscored by its fleet of icebreakers, the largest in the world, which includes a total of around 50 icebreakers, seven of which are active nuclear-powered icebreakers and several more under construction. These icebreakers are essential for ensuring year-round navigability of the Northern Sea Route and for supporting military and commercial activities in the increasingly accessible Arctic waters.

The United States' maritime capabilities in the Arctic are much more limited compared to Russia's, with a focus primarily on its Coast Guard and Navy assets based in Alaska. The U.S. Coast Guard plays a critical role in maintaining Arctic maritime presence, particularly through its icebreaking fleet. However, this fleet is significantly smaller than Russia's, consisting of only two operational icebreakers, including the Polar Star, the nation's only heavy icebreaker, and the medium icebreaker Healy. The limited number of icebreakers poses challenges for the U.S. in ensuring year-round access and operability in the harsh Arctic environment. In addition to its ice breaking fleet, the U.S. Navy maintains a presence in the Arctic through its surface vessels and submarines, although these operations are less frequent and extensive than those of Russia. U.S. submarines, including the Virginia-class, are capable of operating in the Arctic, providing strategic capabilities such as under-ice patrols and surveillance. However, the U.S. lacks a dedicated Arctic surface fleet, relying instead on periodic deployments and exercises to maintain a presence in the region. The U.S. has recognized the growing importance of the Arctic in global security and has begun efforts to enhance its maritime capabilities in the region. This includes plans to acquire new icebreakers, improve infrastructure, and increase joint exercises with allies to ensure a more consistent and capable maritime presence. Despite these efforts, the U.S. still faces significant challenges in matching the extensive and well-established maritime capabilities of Russia in the Arctic.

4.2.2: Air Capabilities

US Air capabilities in the Arctic are also inferior to those of its Russian counterpart, if not technologically, certainly in numbers. The USCG, USN and USAF all contribute to US Arctic air capabilities. The USCG operates multiple aircraft in the Arctic, including the HC-130J for patrol and surveillance, and the MH-60T and MH-65C/D helicopters for recovery and rotary-wing operations. In Alaska, the USAF operates from Joint Base Elmendorf-Richardson (JBER) and

Eielson Air Force Base (AFB), with a significant presence of 54 fifth-generation F-22 Raptor fighters, complemented by mobility, logistics, and refueling aircraft.⁵⁰ At JBER, the Alaska National Guard provides search and rescue (SAR) capabilities with inflight refuelable H-60 helicopters supported by C-130 tankers from the same squadron. The USAF also maintains six F-35 Lightning II fighters at Eielson AFB, with plans to expand its Alaskan capabilities significantly in the next decade by adding 48 more F-35As and eight C-130Js for the Air National Guard and Reserve.⁵¹ The USAF is also upgrading infrastructure at Naval Air Station Keflavik in Iceland, which includes the deployment of KC-135 refueling tankers. The USN enhances this capability by deploying P-8 Poseidon aircraft from Keflavik and exploring operations in Alaska, in addition to its ability to deploy aircraft carrier strike groups near the Arctic.

Russia's Arctic air capabilities are a cornerstone of its defense strategy in the region, with a wide array of advanced aircraft, missiles, and air defense systems spread across strategic bases. The 45th Air Force and Air Defence Army oversees much of Russia's Arctic air operations, including the deployment of Tu-22M3 "Backfire" bombers equipped with Kh-32 supersonic and Kh-47M2 Kinzhal hypersonic anti-ship missiles, providing Russia with long-range strike capabilities. The 98th Separate Mixed Aviation Regiment operates MiG-31BM interceptors in an air defense role and some MiG-31K variants armed with Kinzhal missiles for anti-ship missions. This regiment is transitioning from Su-24M bombers to the more advanced Su-34 Fullback, which will carry Kh-35U anti-ship missiles, enhancing its maritime strike capabilities. In addition, Russia's 100th Independent Shipborne Fighter Aviation Regiment and 279th Shipborne Fighter Aviation Regiment at Severomorsk operate a mix of Su-33s, Su-27s, and MiG-29Ks, all of which are capable of both air superiority and ground attack missions, with some aircraft deployed for interim land-based operations at Rogachevo Airfield in the Arctic. Russia has also heavily invested in air defense across the Arctic, as part of its bastion defense concept, deploying advanced S-400 and S-300 surface-to-air missile systems at critical locations such as Alexandra Land, Kotelny Island, Wrangel Island, and Tiksi. These missile systems provide a formidable shield against both air and missile threats, securing Russia's northern airspace and vital Arctic interests. Additional assets include the 73rd Independent Air Squadron with Tu-142 Maritime Patrol and anti-submarine warfare (ASW) aircraft, as well as the 830th Independent Shipborne Anti-Submarine Helicopter Regiment, which operates Ka-27 and Ka-29 helicopters for ASW and attack roles. The breadth and depth of Russia's Arctic air capabilities enable the country to perform effective surveillance, project power, and exercise strategic deterrence against any threats to its Arctic interests.

In the Air domain, the U.S. has a technological advantage, with its stealthy fifth-generation fighters (F-22 and F-35), which are unmatched in terms of air-to-air combat, advanced sensors, and strike capabilities. However, Russia has superior Arctic infrastructure, a stronger geographic presence in the region, and specialized long-range bombers equipped with advanced anti-ship missiles, making it highly capable in defending and projecting power across the Arctic.

⁵⁰ Abbie Tingstad et. al, "Report on the Arctic Capabilities of the US Armed Forces," *RAND*, 2023, 23.

⁵¹ Ibid

4.2.2: Intelligence, Surveillance and Reconnaissance

Intelligence, surveillance and reconnaissance capabilities, support states' military operations, through the collection, analysis and dissemination of information about the enemy's forces and activities. ISR systems are crucial for monitoring activities and deployments, identifying potential threats and aiding decision-making. As mentioned in section III, having advanced ISR systems is particularly important in an air-sea combat, where the strategic advantage is given by a state's ability to successfully detect, track, and target the enemy's assets. ISR capabilities span various domains, from space, to air, sea and ground, and they are often dual purpose, meaning that they can carry out both offensive and defensive operations.

For the United States, domain awareness in the Arctic region comes primarily from three sources: USCG icebreakers, other surface assets and aircraft and satellites. The USCG represents the primary eyes and ears of the US, providing some of the best domain-awareness information through its regular cutter patrols. Aircraft such as the P-8 Poseidon, stationed in Alaska and rotating through Keflavik in Iceland, significantly enhance maritime surveillance, and can also operate in anti-submarine and anti-surface warfare. Unmanned aerial vehicles are also used by various Arctic states to collect intelligence. Denmark, Canada and the US have purchased surveillance drones, but the high winds, low temperatures, winter darkness and limited satellite communications in the far north mean that few drones currently on the market are able to operate in the Arctic environment. Russia's Arctic UAV program has so far been the most promising in part because of the extensive network of drone bases throughout the Russian Arctic that support maintenance and forward operations. Moscow's ZALA UAV series constitutes large drones capable of collecting images from more than 100 km away.⁵²

On the ground, ISR systems encompass early warning systems, ground-surveillance radars and other sensors. The Alaska Radar System comprises 15 long-range and short-range radars, three of which belong to the North Warning System, a joint US-Canadian early-warning system for the atmospheric air defense of North America. At Eareckson AS, the US Space Force hosts another missile defense radar, and in the Pituffik Air Base in northwest Greenland they have a ballistic-missile early-warning and space domain-awareness system. While radars and early warning systems are used to aid offensive operations, they are generally used for defensive purposes. During the Cold War, the Soviet Union also constructed a series of radar stations along the country's northern coast – the Missile Attack Warning System (SPRN) – which remains active today. Additionally, Russia has deployed Rezonans-N radar systems to bases on Novaya Zemlya and at Indiga, and they may be able to detect hypersonic missiles and stealth planes. Sopka-2 radar systems are instead deployed at the Temp, Nagurskoye and Rogachevo bases.⁵³

⁵² Kathryn Urban, "A New (Cold) Front in Polar Intelligence? Trends and Implications of Technology-Enabled Monitoring in the Arctic," *Journal of Science Policy & Governance*, January 11, 2021, 4, <https://doi.org/10.38126/jspg190111>.

⁵³ Colin Wall and Njord Wegge, "The Russian Arctic Threat: Consequences of the Ukraine War," *Www.csis.org*, January 25, 2023, <https://www.csis.org/analysis/russian-arctic-threat-consequences-ukraine-war>.

Arctic submarines played a very strategic role during the Cold War for intelligence collection and monitoring, in addition to warfighting. Technological advancements made these vessels so quiet that they became difficult to detect, so submarines would often come into direct confrontation in the Arctic. NATO states would routinely patrol the Greenland-Iceland-United Kingdom (GIUK) gap, and Soviet submarines would always try to observe NATO naval forces in their regular under-ice patrols. Although the frequency of submarine patrols has declined substantially in the 21-st century, NATO members continue to perform submarine surveillance in the GIUK Gap, which is crucial to maintaining NATO's transatlantic lines of communication. Russian submarines also routinely monitor undersea communications cables for intelligence gathering and the country's submarine program consistently receives funding to support conventional tracking activities and new deep sea data gathering capabilities.⁵⁴ Modern underwater sensors are able to detect the acoustic signatures and noises produced by submarines, and are therefore a key strategic ISR system. Among the most innovative systems in the maritime domain there are sensor-enhanced buoys and uncrewed undersea vehicles such as drone submarines.⁵⁵

In the space domain, countries often utilize commercial satellites and open source data to gather information on the enemy's activities. In the Arctic, satellites become extremely important because of the harsh environmental conditions that characterize this region, making other types of surveillance activities a lot more costly. The USSF acquires, operates and sustains the Global Positioning System (GPS), providing 24/7 global coverage and highly accurate position, velocity, and timing data to widely deployed GPS user equipment. Although GPS service is not optimized for the poles, access to GPS signals enhances real-time battlespace awareness for command and control (C2), synchronizes communications, and enables all forms of precision operations, such as target location, weapon delivery, SAR, and logistical support to U.S. and allied users. Another main source of Arctic intelligence is Canada's satellite, RADARSAT-II, which offers nearly constant feed on Russian behavior. US space assets controlled from its base in Alaska and Greenland, also periodically survey the region with imaging passes over the Far North made every thirty minutes.⁵⁶ The Russian alternative to GPS is known as GLONASS, which stands for Global Navigation Satellite System. GLONASS signal availability tends to be stronger in northern latitudes, hence providing better coverage of the Arctic and the North Pole regions compared to GPS. Russia also has other space-based ISR capabilities specific to the Arctic – Meridian-M communications satellites and an Arktika-M weather satellite for navigation support.⁵⁷

⁵⁴ Kathryn Urban, "A New (Cold) Front in Polar Intelligence? Trends and Implications of Technology-Enabled Monitoring in the Arctic," *Journal of Science Policy & Governance*, January 11, 2021, 4, <https://doi.org/10.38126/jspg190111>.

⁵⁵ Ibid

⁵⁶ Ibid, 5

⁵⁷ Colin Wall and Njord Wegge, "The Russian Arctic Threat: Consequences of the Ukraine War," *Www.csis.org*, January 25, 2023, 6
<https://www.csis.org/analysis/russian-arctic-threat-consequences-ukraine-war>.

V. Three Policy Options to Respond to Russia's Militarization

Now that I've laid out each country's interests and capabilities to operate in the Arctic region, I can evaluate under what conditions each policy option would be optimal for the United States in the Arctic. As a reminder for the reader, I have extrapolated the three policy options that I will be discussing from Glaser's optimal vs suboptimal theory of arming. Two of them, the Arms Build-up and the Self Restraint policies are the direct result of Glaser's theory, while the third one, the ISR/Domain Awareness Policy, is an additional option that I propose as a third way that might better suit current Arctic conditions. In this section, I will describe the conditions under which each policy would be the optimal policy for the United States to pursue in response to Russia's threats in the Arctic. This discussion builds on the previous sections of the paper, especially sections III and IV, to provide a better understanding of which option best applies to current Arctic conditions in a way that maximizes US interests in the region.

Policy Option 1: Arms Build-up Policy

This policy option entails boosting the US military capabilities in the Arctic through an intense build-up of Arctic-specific assets. The benefits of pursuing this policy are associated with increased general deterrence and greater ability to protect the US homeland and the US allies from a potential Russian attack in the region. The costs associated with such a policy have to do with the potential reduction of crisis stability and the monetary costs associated with building the required infrastructure. So, what conditions would make this policy the ideal policy for the US to pursue?

First and foremost, the probability of a war breaking out in this theater has to be moderately high in order to justify both the economic resources that the US would have to invest in new Arctic assets, and the escalatory risk associated with building arms. Building up arms when the probability of warfighting is low, might simply increase the likelihood of conflict escalation. Determining whether the probability of war onset is high depends on a variety of factors, including the existence of highly disputed territories or interests, the capability for at least one state to effectively win in a war, and the exhaustion of all other peaceful forms of bargaining, such as diplomacy or sanctions. The US would therefore need to have highly significant interests in the region to make the potential for crisis escalation worth pursuing and have already tried deterring a potential Russian offense through legal, de-escalatory means.

If those options have been exhausted, then building arms might be the US next best option. In that case, the US needs to have the resources necessary to invest in building offensive Arctic capabilities. In other words, the US would need to have a power advantage, where power is the ratio of a state's resources that can be converted into military assets (from Glaser's model). Moreover, for an Arms Race Policy to be optimal, the offense-defense balance (ODB) would need to favor offense. When ODB favors offense, it means that it is easier, less costly and overall more advantageous for a state to conduct successful offensive operations. In such a scenario it would be desirable for the US to have quantitatively and qualitatively comparable military capabilities to Russia to guarantee the US the ability to launch an offensive operation against its opponent, if needed. Hence, the US would be better off building up arms.

Another key variable to consider when deciding whether it is optimal for the US to pursue an arms race policy, is to determine whether Russia is a greedy state or a

security-seeking state. According to the deterrence model, conflict is the result of greedy adversaries: states interested in expansion for non-security reasons. If the US believes that Russia is a greedy state, then building up arms could enhance deterrence and disincentivize Russia from launching an attack. This is true only if the marginal value of deterrence is high, meaning that even a small increase in a state's military capabilities significantly enhances its ability to deter adversaries. However, if Russia is so greedy that it is willing to put its homeland at risk, then the US decision to build up arms might still not deter a Russian attack.

To sum up, an Arms Race Policy would be the optimal US policy decision in the context of the Arctic if the following conditions are met:

- The probability of warfighting in this theater is high
- The US has vital interests in the region and has exhausted all peaceful way to de-escalate the crisis
- The US has a power advantage and the ability to outbuild Russia in offensive capabilities
- Russia is a greedy state that threatens US interests and security
- The offense distinguishability is low or offense-defense balance favors offense
- The marginal value of deterrence is high

It's important to notice that if only one of the conditions is met, this policy is NOT the preferred policy option for the US. Given how high the cost of warfighting is, the US would need to face the majority of these conditions for this policy to be optimal. This is particularly relevant in the case of an arms race policy and does not apply to policy options 2 and 3.

Policy Option 2: Self-Restraint Policy

A Self-Restraint Policy calls for the maintenance of the current status quo in the Arctic, with Russia being quantitatively superior to the US when it comes to military assets in the region. The main benefits of such a policy are: communicating benign motives to the opponent, containing crisis escalation, and allocating resources to places that are of higher priority for the US. The downsides of this policy primarily concern the consequences of letting Russia be the dominant Arctic power. Such a scenario could make the US and its allies more vulnerable to a potential Russian attack, and unable to deter or effectively respond to it.

The ideal conditions for a Self-Restraint Policy to be successful somewhat mirror those that make an Arms Build-up Policy ideal, but in reverse. Hence, for policy option 2 to be optimal, the US should perceive the probability of war onset in the Arctic to be relatively small, at least in the near future. This doesn't mean that the probability of a war to *ever* start in the Arctic has to be low, but that such a scenario is perceived to be a future possibility and not a current urgency. Moreover, pursuing this policy presupposes that the US interests in the region are not vital, and that there are better, less costly ways for the US to deal with Russian threats.

Another reason not to build up arms is associated with power. If the US does NOT have a power advantage, meaning that the costs of acquiring the capabilities required to launch or defend from an attack are excessive, then there is a high probability that it will fall behind in an arms race. In this case, it would be a pointless pursuit to start building up arms knowing that Russia has a greater chance of winning the arms race. In the event that the US does have the

resources to invest in Arctic military assets and potentially win in an arms race, Self-Restraint might still be the optimal choice if the US perceives Russia to be a security-seeking state. In other words, if the US believes that Russia's militarization of the Arctic is motivated primarily by fear of encirclement and NATO enlargement, even more so after the admission of Finland and Sweden into NATO, then choosing not to build arms would signal benign motives. If the US decided to build up its military despite the fact that Russia is security-seeking, then, according to the spiral model, that could exacerbate the security dilemma and increase the likelihood of conflict escalation. The successful communication of benign motives implies that Russia is reassured and convinced that the US does not intend to attack Russia or threaten its interests in the Arctic. Failing to effectively communicate US intentions would cause Russia to mistrust the US and continue its militarization process as a preemptive measure.

Ultimately, the Self-Restraint Policy option would be optimal if the offense-defense balance favored defense over offense. When defense offsets offense, it means that a state's defensive capabilities are sufficiently strong to neutralize or counteract the advantages of the opponent's offensive actions. In other words, even if the adversary has the capabilities to launch a successful offensive operation, the defending state's defensive measures are robust enough to prevent the offensive side from achieving its objectives or to make the offensive extremely costly and difficult. In this scenario, the US should either already own the defensive capabilities required by the ODB, or it should be easily capable of acquiring them if the likelihood of conflict was to increase significantly.

In short, a Self-Restraint Policy in the Arctic would be the ideal policy option if the following condition were satisfied:

- The probability of warfighting in this theater is currently low
- The US doesn't have a power advantage and the probability of falling behind in an arms race with Russia is high
- The US interests in the Arctic are not vital and other theaters are more valuable
- Russia is a security seeking state and building arms might trigger a security dilemma
- The US is capable of signaling benign motives and appease Russia's security concerns
- The offense-defense balance favors defense

While policy option 1 requires most of the conditions to be met for it to be the ideal policy option, for policy option 2 to be optimal it is sufficient that one of the above conditions is present. This is because building up arms when one of those conditions applies would lead to suboptimal arming and might result in undesirable outcomes.

Policy Option 3: Domain Awareness Policy

Although Glaser's theoretical framework primarily focuses on arming vs self-restraint, it does mention how monitoring capabilities can play a role in preventing conflict escalation. Glaser explains that the better the state's monitoring capabilities, the smaller the lead its adversary could achieve by launching a buildup first and, therefore, the smaller the danger in

cooperating.⁵⁸ Having appropriate domain awareness capabilities would be even more critical in the context of the Arctic, where warfighting would happen primarily in the sea and air spaces, and guided missiles would be the primary weapons used against the adversary. In this context ISR becomes the key means through which a state can accurately detect an adversary's attack and accurately strike the opponent's strategic targets. The benefits of this policy are therefore linked with an enhanced situational awareness and ability to track the adversary's assets and strategic infrastructure, facilitation of strategic communication, more effective decision-making, and reduced opportunistic miscalculation. On the other hand, ISR systems are very costly to produce, especially if they need to be able to endure Arctic extreme weather conditions, and depending on the nature of the system, ISR capabilities might also escalate tensions between the US and Russia.

The primary condition for a Domain Awareness Policy option is therefore already in place, namely the fact that Arctic warfighting requires ISR for precision strikes. While the offensive use of ISR systems becomes critical only in the event of a war onset, their defensive capability remains critical up until the start of a war, and during the war. Crucially, investing in domain awareness systems is ideal when there is a non-negligible probability of conflict escalation in the Arctic, but it is unclear if, when and to what extent conflict would escalate. This would be the case if Russia's activities in the region are limited to minor disputes, hybrid threats, and power projection operations. Such actions lay below the threshold of war initiation, so the US does not need to build up arms to respond to such threats, but it still requires strict monitoring and situational awareness to evaluate the severity of the actions and ensure that law and order can be restored and peace maintained in the region. In this scenario, US interests in the region need to be substantial but not vital to its survival, preferring a laid back approach to the active pursuit of arms building.

This policy option is also ideal both when the adversary is security seeking or greedy. If the opponent is greedy, then ISR building allows a state to monitor the activities performed by the greedy state and it can help calibrate the amount of arms it needs to defend itself from an attack by the greedy state. If the adversary is security-seeking, a domain awareness build-up would still be ideal because it communicates to the opponent the state's willingness not to engage in an arms race, while still recognizing the need to oversee its actions. Hence, if the US believes that Russia is security-seeking and that an arms build-up would significantly increase the likelihood of a security dilemma, then an ISR policy paired with a self-restraint policy would be preferable. In this case, the production and deployment of ISR should be very explicit to ensure that Russia perceives such action to be benign in nature. It probably goes without saying, but such a policy requires that the US has the resources needed to build such expensive infrastructure.

In terms of ODB, if offense and defense are distinguishable, then an ISR Policy would be ideal if offense favors defense, because the usefulness of building ISR is greater for both the offensive and the defensive side if ODB favors offense. This is because the ability to detect, track and strike first can be decisive when both sides have an offensive advantage. ODB favors defense, the usefulness of building ISR is low for both sides (See figure 1). However, if offensive and defensive capabilities are indistinguishable, then both states are better off pursuing an ISR

⁵⁸ Charles L. Glaser, "When Are Arms Races Dangerous? Rational versus Suboptimal Arming," *International Security* 28, no. 4 (April 2004): 59, <https://doi.org/10.1162/0162288041588313>.

policy, because the sole fact of not knowing the nature of the adversary's capabilities makes it ideal to build up domain awareness systems.

Figure 1

Usefulness of ISR	Offense favorable	Defense favorable
Offensive side (Russia)	High	Low
Defensive side (US)	High	Low

To sum up, policy option 3 is ideal under the following conditions:

- There is a probability for conflict escalation in the Arctic but it is unclear if, when and to what extent conflict would escalate
- The US has some interests in the Arctic region, but they are not vital to US survival
- The US has the resources and power to build ISR capabilities
- The adversary is either security seeking or greedy
- The offense-defense balance favors offense if offense and defense are distinguishable
- Offense and defense are indistinguishable

VI. What Policy Option Better Applies to Current Arctic Conditions?

Now that I have delineated the conditions under which each policy would be optimal, I must determine which policy option best applies to the current Arctic conditions. In order to do so, I will break down current conditions into: conflict escalation probability, states' interests in the region, and material and informational variables (power, ODB, and greedy vs security-seeking motives).

Conflict Escalation Probability

Currently, the probability of conflict escalation between the US and Russia in the Arctic is low. Not only is Russia too busy fighting a war in Ukraine, but it has also not exhausted the peaceful, less-costly means to achieve its Arctic interests. On the contrary, in 2023, the UN commission on the Laws of the Sea approved Russia's continental claims submitted in 2015 and ratified in 2021. The commission found that most of the extended continental shelf claims made by Russia were legitimate (1.7 million square kilometers of seabed), and only 300,000 square kilometers did not pass the scientific threshold for recommendation.⁵⁹ Although Russia is not satisfied with this decision and has already submitted further scientific evidence to claim the remaining 300,000 square kilometers, the commission's recommendation in favor of Russia, is already a significant gain for Moscow. If Russia is able to achieve its Arctic objectives through

⁵⁹ Elizabeth Buchanan, "Russia's Gains in the Great Arctic Race," War on the Rocks, May 4, 2023, <https://warontherocks.com/2023/05/russias-gains-in-the-great-arctic-race/>.

legal means, then the likelihood of conflict escalation is significantly reduced, especially given how costly wars are. While Russia's adherence to the international legal framework seems promising in theory, its actions in Ukraine demonstrate that the country might easily experience a change of heart. This would be true especially in the event of any Western push backs against the commission's ruling. The fact that the United States has not ratified UNCLOS might raise some concerns regarding the potential for US-Russian confrontations over the legal status of the Arctic territory and waterways, but the probability that such disputes would escalate into a conflict remain low, as peaceful negotiations can still be effective in settling down such disputes. Russia has however utilized threatening measures to signal its commitments to the Arctic and its ability to use hard power if necessary. Some of these measures include hybrid threats and electronic warfare, such as the tampering of undersea sensors and fiber optic cables linking Norway and the Svalbard archipelago, air and sea patrols off the coasts of Canada and Norway, GPS jamming, and increased missile tests and military exercises that signal Russia's readiness to use arms. Such threats cannot be ignored. Even though they lay below the threshold of warfighting and do not pose a real risk of conflict at the moment, they require constant monitoring to ensure that they don't escalate further in the future.

How vital are states' interests?

Section IV was dedicated to analyzing Arctic states' interests in the region, focusing primarily on the economic and strategic objectives of the United States and Russia. That section set the stage to now evaluate whether the US and Russia perceive the Arctic to be a vital region for the development of their military and economic goals. The logic is that the greater the value that a state allocates to the region, the greater their willingness to use military power to back up those claims. The use of force becomes especially relevant when all other options of crisis bargaining have failed. According to Jonathan Markowitz's book, the Perils of Plenty, resource-dependent states like Russia and Norway are more likely to use power projection as a coercive means to maintain access to the region's resources, while production-oriented states like the United States and Denmark are less likely to fight for newly discovered resources, because their economic growth does not depend on them. Based on the Rent-Addiction theory, the Arctic region is not a vital location for the United States, at least from an economic perspective, while it is extremely valuable for the resource-dependent Russian Federation. Russia's interests in the Arctic are overall more vital than US interests. This is because Russia's interests revolve around the future economic development of the country, the country's Great Power status and the protection of its nuclear second-strike capabilities. On the other hand, the US main objectives are to ensure that the Arctic remains peaceful, that the Russian and Chinese increased activities in the region don't disrupt key lines of communication between North America and Europe, and that strategic maritime chokepoints remain open to US transit. Hence, while the US has some strategic stakes in the region, the Arctic territorial disputes don't directly threaten the US future economic survival and Great Power status. On the other hand, the Arctic is vital for Russia's future as a Great Power.

While the Arctic region is far more crucial for the future economic development of Russia than it is for the United States, I believe that the region becomes an invaluable location for the US insofar as it is vital for the Russian Federation. This argument echoes the offensive realism theory, according to which great powers are power maximizing entities that constantly strive to

dominate the international system. Given that Russia is a revisionist power that seeks to use the Arctic as a platform to secure its Great Power status, the United States will want to prevent Russia from dominating in the Arctic geopolitical context and will seek to stop Russian expansion of influence in the region. Before Russia started investing in modernizing its Arctic capabilities, the United States had little interest in the Arctic; it wasn't until 2013 that the US released the first Arctic Strategy since the Cold War, and only in recent years the region became one of US national priorities. The US renewed interest in the Arctic is therefore a direct result of Russia's efforts to expand its influence in the region. Hence, even though the US interests in the region are somewhat limited, they become more vital as a result of Russia's economic and strategic ambitions in the region.

Material Variables

According to Glaser's framework there are two material variables: power and offense-defense balance. To measure power, which is the ratio of resources that a state can convert into military assets, I decided to use the states' defense budgets adjusted for purchasing power parity. In 2023, the US military expenditure on the defense sector was approximately \$854 billion USD, compared to Russia's \$121 billion USD.⁶⁰ This suggests that the US defense budget is about seven times larger than Russia's budget. While it would be an oversimplification to conclude that the US is seven times as powerful as Russia, this staggering difference in military spending between the two countries suggests that the US has an overall power advantage, as it has more resources that it can convert into military assets compared to Russia. Then why is Russia militarily more powerful than the United States when it comes to the High North? In short, this is because having the power to build military capabilities does not translate to having the willingness to do so. Russia has prioritized the Arctic region throughout history and it has slowly but consistently invested in the modernization of Arctic bases and military forces. The United States, on the other hand, has prioritized other geopolitical regions in the past few decades, such as the Middle East and Far East Asia, neglecting the Arctic, which had become a peaceful region following the end of the Cold War. The limited economic interests in the region, paired with the higher geopolitical priorities in the Far East like Taiwan and the Philippines, make the US reluctant to invest more in Arctic-specific assets. However, as Russia and China continue to expand their influence in the Arctic, the US will need to invest greater resources in the High North, so it is likely that the US will obtain greater power in the region in the near future. As of right now, the US has an overall power advantage, but Russia is more powerful when it comes to the Arctic.

The offense-defense balance is much harder to measure. A rigorous analysis would require comparing the costs of forces required by Russia to win an attack in the Arctic and for the US to defend itself from such an attack. Doing so quantitatively doesn't only include comparing the amount of assets each country has in the region, but also the accuracy of the missiles they launch, their range, and the amount of missiles each asset can carry. This analysis

⁶⁰ Lowy Institute, "Military Expenditure, Defence Sector PPP Data - Lowy Institute Asia Power Index," Lowy Institute Asia Power Index 2021, n.d., <https://power.lowyinstitute.org/data/military-capability/defence-spending/military-expenditure-defence-sector-ppp/>.

would be done primarily for conventional sea and air forces, given that neither country can technically use nuclear weapons to strike first. This net-assessment type of analysis is tedious and often requires access to sources that are not open-source, hence I will be justifying my ODB conclusion qualitatively, referring back to the capabilities section in part IV.

Currently, the ODB in the Arctic favors defense for the United States for several reasons. Firstly, the US currently lacks some of the most fundamental assets to launch a successful offensive operation in the High North. The US only has two operational icebreakers, which are used for both Arctic and Antarctic missions. Without an icebreaker fleet, it's impossible for the US to deploy its powerful destroyers and other warships to the region, making it difficult to win in an attack against Russia, which currently owns about 50 icebreakers. Moreover, the United States doesn't have naval assets readily deployed in the Arctic region, and would have to remove them from other geographic locations, such as Taiwan, to launch a potential attack against Russia in the Arctic. The opportunity cost of doing so is very high, considering that the probability of conflict escalation in the region is still low. US aircraft deployed in Alaska are composed primarily of F-22 Raptors and F-35 Lightning II, both of which have the capability of striking a Russian destroyer, even though the F-22 is not optimized for it and is mainly employed in air-to-air combat. The F-35, equipped with precision-guided munitions is much more versatile and better suited for maritime strike missions, but there are only 6 of these combat aircraft currently deployed in Alaska, making it difficult to carry out an offensive operation against Russia depending on current US air forces. Hence, the costs of acquiring the capabilities to launch a successful offensive operation in the Arctic is very high. According to an estimate I made using the Defense Simulator tool, the US would have to invest an additional \$80 billion in new Arctic military assets over the next five years to have Arctic capabilities comparable to Russia's.⁶¹

Secondly, while the US has increased the number and frequency of military exercises in the High North in recent years, the ability of soldiers to operate in the harsh conditions of the Arctic environment is significantly lower than Russia's, which routinely operates along the Siberian coast. The extreme weather also creates a natural barrier to offensive operations, as moving large forces, maintaining supply lines, and sustaining search and rescue operations becomes extremely difficult. Due to the long distances, both air and naval support must operate over vast and remote areas with limited infrastructure, further complicating logistics and reducing operational efficiency. The lack of established bases, fuel depots, and reliable communication networks in the Arctic heightens the challenge for US forces, which would need to rely heavily on pre-positioning supplies or forward operating bases. This logistical strain not only increases the cost and complexity of offensive operations but also makes it difficult to sustain them over prolonged periods of time.

Acquiring defensive capabilities is therefore less costly and more convenient for the United States, which can also rely on the capabilities of its NATO allies if a Russian threat in the Arctic becomes more realistic. By relying on this alliance network, the US can avoid the costly and complex process of building a full-scale offensive force tailored to Arctic operations. Instead, it can focus on bolstering the joint defense infrastructure through increased

⁶¹ "Defense Futures Simulator," Cloudfront.net, 2024, <https://drepjolqp87hn.cloudfront.net/drkk2/dashboard/navy>.

collaboration, joint exercises, and intelligence sharing. This makes defense more efficient and ensures a rapid, coordinated response if a Russian threat in the Arctic escalates.

Informational Variables

According to Glaser's theory, a state can be classified as either greedy or security-seeking, though distinguishing between the two is often difficult. Some might argue that Russia is a greedy state, as it has sought to expand its power and influence by annexing territory belonging to other sovereign nations. This was evident in its occupation of Abkhazia and South Ossetia during the 2008 Russo-Georgian War, the annexation of Crimea in 2014, and its invasion of Eastern Ukraine in 2022. If one believes that Russia's actions stem from motives beyond security—such as a desire to expand influence or gain resources—then it fits the description of a greedy state. On the other hand, if one views these actions as efforts to create a buffer zone for self-defense against NATO's expansion, fearing encirclement by adversaries, then Russia could be seen as security-seeking.

The United States likely perceives Russia as a greedy state. Russia is often regarded as a revisionist power, seeking to reshape the current international system dominated by the West. Driven by ambitions of influence and power, Russia probably views the Arctic as an economic avenue, aiming to exploit its resources to enrich itself and fund future military endeavors. Annexing territories that have either economic, political or strategic value is therefore one way to accomplish its revisionist goals. From its invasions of Georgia and Ukraine to its hybrid tactics in the Arctic and other acts of subversion, Russia projects itself as an adversary to the US, fitting the label of a greedy state. Whether these perceptions are fully accurate or not is less important for determining the optimal US policy in the Arctic. What truly matters is the US's belief that Russia is a greedy state. Given that the US most likely holds the view that Russia is a greedy state, maintaining the current status quo in the Arctic would be suboptimal. A greedy state must be deterred and contained, requiring greater operability and enhanced domain awareness in the region to counter Russia's ambitions.

So, what's the ideal policy option, given current Arctic circumstances?

Given all the above considerations, the policy option that best responds to US needs in the Arctic is policy option 3, the domain awareness policy. The low probability of conflict escalation suggests that building up arms would be a suboptimal decision. US interests in the Arctic are limited to ensuring that the region remains peaceful and that its homeland is safe from a potential Russian attack. Hence, with a low risk of conflict escalation, the region is not strictly vital to US survival. Alaska already serves the defensive purposes needed by the US to protect its homeland, and the US has little interest in the economic opportunities available in the region, as extracting resources from the Arctic requires expensive investments and the US is predominantly a production-oriented state. Russia's increased investment in militarizing the region must, however, be constantly monitored, as it might eventually escalate to a level above that of hybrid warfare. This requires advanced domain awareness capabilities, from new ice-breakers to cold-enduring drones, and more advanced radar systems.

Given that the US considers Russia to be a greedy state, who might eventually pose a real threat to US security or that of its allies, it is even more crucial that the US maintains high levels of domain awareness in the region, to be able to promptly respond to Russia's actions, if

needed. Maintaining the status quo would therefore also be suboptimal, because, as Russia continues to increase its military investment, the US will need to improve its surveillance capabilities to better detect potential threats and malign activities. Investing in better ISR systems is therefore of key strategic importance in this moment of uncertainty surrounding Russia's intentions.

As we have seen, the offense-defense balance currently favors defense for the United States, meaning that it is easier, less costly and more effective to defend territory than it is to launch an offensive operation. ISR is generally most useful when ODB favors offense for both sides. However, since offensive and defensive capabilities are not distinguishable in air-sea warfare, given that the emphasis in this type of warfare is on preemptive strikes, ISR capabilities become extremely important. Not only is information superiority critical in an offensive scenario, to be able to quickly and effectively strike the enemy's naval and air forces, but it is also crucial in a defensive scenario to react and respond quickly to an attack. While the US is better off pursuing a defensive strategy rather than an offensive one, the indistinguishability of offensive and defensive capabilities in the Arctic makes ISR particularly useful. This makes an ISR policy the optimal policy for the US to pursue in the region.

VII. Conclusion

My analysis concluded that the optimal policy for the US to pursue in the Arctic is a domain awareness policy. This conclusion was the result of an in-depth evaluation of the US and Russia's stakes in the region, as well as their military capabilities, considered within the theoretical framework of "optimal vs suboptimal arming" developed by Charles Glaser. Despite being the optimal approach given the current geopolitical conditions in the Arctic, deciding how to operationalize it and what ISR systems to invest in requires further research. ISR systems can be both stealth and non-stealth, some are capable of detection only while others can both detect and track the opponent's assets. Depending on the nature of the system purchased, an ISR policy can either escalate or de-escalate tensions in the Arctic. For instance, imagery satellites are usually stabilizing systems, as they are non-stealth and capable of detection only, helping to reduce mutual miscalculations and enabling the adversary's contingency countermeasures. On the other hand, ISR systems such as stealth unmanned reconnaissance aircraft might reduce crisis stability, as they are not only able to detect targets, but also to track them, significantly increasing the adversary's vulnerability and disabling the adversary's contingency measures in case of an attack. Carefully weighing the advantages and disadvantages of different domain awareness technologies is therefore of extreme importance and further research should dive deeper into what systems are more suitable for the achievement of US interests in the region.

Through this paper I aimed to evaluate how the US shall respond to Russia's increased militarization of the Arctic, using a broadly accepted and recognized framework. The decision to apply IR theory to the Arctic case study is innovative on its own, as most papers and reports written on the topic take a more hands-on approach that ignores some of the more academia-style questions, such as whether building arms might escalate tensions by triggering a security dilemma. By choosing to use a rigorous and theory-heavy methodology, I hope to have made a contribution to the debate on whether the Arctic region will become a new theater

of war in the near future, and what the US should do to prevent it. I hope to have provided a new perspective on how to think about this topic, and to have demonstrated how difficult it is to answer a seemingly simple question: to arm or not to arm? I recognize that this paper has some limitations, in part due to the difficulties associated with accessing sensitive data on the exact number of missiles each country has and can use in the Arctic region, and in part because Glaser's theoretical framework might better apply to land warfare motivated by territorial gains, rather than a hypothetical escalation of tensions in the remote Arctic region. However, insofar as my analysis provides a useful lens for understanding the strategic dynamics of arms competition as they apply to the Arctic, it still offers valuable insights into the implications of a military buildup in the region. While conflict escalation remains a relatively low probability, this paper may serve as a theoretical foundation for policymakers and scholars alike as they decide how to balance general deterrence and diplomatic restraint in the face of Russia's militarization of the region. This paper proposes a domain-awareness policy as a middle ground between full-scale militarization and passive restraint. By prioritizing advanced surveillance and monitoring technologies, the US can maintain a strong defensive posture and an enhanced situational awareness without directly contributing to an arms race that could further destabilize the region. This policy appears therefore to be the most promising and effective solution to Russia's threats in the High North, enabling the US to respond swiftly and proportionally to potential aggression, ensuring both regional stability and the protection of vital strategic interests in the Arctic.

Appendix

Northern Fleet (the western Arctic region)

Russian Submarines Northern Fleet		
Submarine class	Missile capacity (per vessel)	Count
Borei Class	16 Bulava SLBMs containing 6 MIRVs 8 x 533mm torpedo tubes 6 x 533mm anti torpedo defense system	2 in the Northern Fleet 1 in the Pacific Fleet 4 under construction
Borei-K-Class	16 Cruise missiles	2 in service 6 on the way
Yasen class multipurpose-class SSGN	10 x torpedo tubes armed with UGST-M heavyweight guided torpedoes and eight vertical launch silos for P-800 Oniks missiles	2 active in NF 1 floated 1 under construction
Delta IV	Carry up to 16 ballistic missiles, each with 4 MIRV; the missiles range is 7,700km	6, but will gradually be replaced by Borei class
Project 945 an 945A (Sierra I and II)	P-100 Oniks anti-ship cruise missiles, RPK-6 Vodopad and TYPe 86R Vodopad anti-ship cruise missiles, and type 40 torpedoes. In lieu of torpedoes, these vessels can carry 42 mines	2 active in the NF
Antey-Class (Oscar I and Oscar II)	4 x 533mm torpedo tubes, two 650mm torpedo tubes, 32 mines	3 active in the Northern Fleet ⁶²
Shchuka-Class (Victor III)	4 x bow torpedo tubes six 533mm tubes with torpedoes. Can carry 36 mines or 24 torpedoes	2 active in the NF
Shchuka B-Class (Akula) nuclear-powered	Carry a total of 40 torpedoes and 1 to 3 surface-to-air missiles; range of 3,000km	3 active in NF 3 to overhaul and modernize

⁶² https://en.wikipedia.org/wiki/Oscar-class_submarine#Units

attack submarines		
Project 877/636 Varshavyanka (NATO: Kilo)	6 x 533mm tubes with 18 torpedoes; 4 Club-S missiles; 24 mines; 8 surface-to-air missiles	4 active in NF
Lana-Class (St. Petersburg)	6 x 533mm torpedo tubes with 16 torpedoes or SSMs; 10 Kalibr/Club missiles	1 in the Baltic
The Poseidon		Over 30 under construction

Russian Surface Warships Northern Fleet		
Surface Warships	Missile Capacity per vessel	Count
Papanin-class Arctic patrol ships		
construction of two more for the Coast Guard commenced in July 2020		
Admiral Kuznetsov aircraft carrier	Armament: 6 x AK-630 AA guns 8 x CADS-N-1 Kashtan CIWS 12 x 8-cell 3K95 Kinzhal SAM VLS RBU-12000 UDAV-1 ASW rocket launchers Aircraft carried: 18 x Su-33 6 x MiG-29K 4 x Ka-31 2 x Ka-27	1
Slava-class cruiser	Missiles: 16 P-1000 Vulkan anti-ship missiles 64 S-300 Fort long-range surface-to-air missiles 40 OSA-M surface-to-air missiles Guns: 1 twin AK-130 dual purpose guns 6 x 1 AK-630 close in weapons systems	1

	Torpedoes: 2 x 12 RBU-6000 anti-submarine mortars 10 533mm torpedo tubes Aircraft carried: 1 Kamov Ka-25 or Kamov Ka-27 helicopter	
Gorshkov-class frigates	Armament: 1 x 130mm Amethyst naval gun 16 cells for Kalibr, Oniks, Zircon anti ship cruise missiles or Otvet anti-submarine missiles 32 more cells for the above missiles 32 cells for surface-to-air missiles 2 x Palash CIWS each with twin Gryazev-Shipunov GSh-6-30 6 barrel 30 mm rotary cannons. 2 x 4 330 mm torpedo tubes for Paket-NK anti-torpedo/anti-submarine torpedoes 2 x 14.5 mm MTPU pedestal machine guns Aircraft carried: 1 x Ka-27 series helicopter	3 active but only 1 currently in the NF
Udaloy I/II-class anti-submarine warfare ships	Udaloy I: 2 x 1 AK-100 100 mm naval guns 8 (2 x 4) SS-N-14 'Silex' anti-submarine/anti-ship missiles 64 (8 x 8) VLS cells for SA-N-9 'Gauntlet' surface-to-air missiles 4 x 6 30 mm AK-630 CIWS 2 x 1 21KM AA guns 2 x 4 533 mm torpedo tubes for Type 53 torpedoes 2 x 12 RBU-6000 anti-submarine rocket launchers Udaloy II: 1 x 2 AK-130 130 mm naval guns 8 (2 x 4) SS-N-22 'Sunburn' anti-ship missiles 64 (8 x 8) VLS cells for SA-N-9 'Gauntlet' surface-to-air missiles 2 x Kashtan CIWS 2 x 4 553 mm torpedo tubes for SS-N-16 'Stallion' anti-submarine missiles 2 x 10 RBU-12000 anti-submarine rocket launchers Aircraft carried: 2 x Ka-27 series helicopters	3 active 1 in refit

Kirov-class battlecruisers	20 x P-700 Granit anti-ship missiles 1 RPK-3 Metel anti-submarine missile 96 x S-300 Fort SAM 48 x S-300F Fort 48x S-300FM Fort-M long range SAM 64 x	1 active 1 in refit
Sovremenny-class destroyer	<u>Guns:</u> 4 AK-130 naval guns 4 x 30mm AK-630 CIWS <u>Missiles:</u> 8 SS-N-22 anti-ship missiles 48 SA-N-7 "Gadfly" surface-to-air missiles <u>Anti-submarine:</u> 2 x 2 533mm torpedo tubes 2 x 6 RBU-1000 300mm anti-submarine rocket launchers <u>Aircraft carried:</u> 1 x Ka-27 series helicopter	1
Landing ships		8
Nanuchka III-class corvette	6 x P-120 or 4 x P-15 anti-ship missiles 16 x Kh-35 anti-ship cruise missiles 20 x 4k33 surface-to-air missiles Twin 57mm Ak-257 gun 1 x 76mm AK-176 gun 1 x 30mm AK-630 gun	1 active
Grisha-class corvette	1 twin SA-N-4 Gecko surface-to-air missile launcher (20 missiles) 1 twin 57mm 70-caliber dual purpose guns 2 RBU-6000 anti-submarine rocket launcher (96 rockets) 2 twin 533mm torpedo tubes 2 depth charge racks Up to 18 mines in place of depth charges	7 Pacific 6 Northern
Ivan Gren-class landing ships	1 x 30mm AK-630-2 gun 2 x 30mm AK-630 guna 2 x 14.5 mm KPV type gun Hangar 2 Ka-27 ASW or Ka-29 transport-attack helicopters on 117 and 135; hangar for 4 helicopters	2 active 1 in

Air force:

US Air Forces in Alaska			
Aircraft	Stealth?	Missile capacity	Count
F-22	Yes	Guns: 1 x 20mm rotary cannon (480 rounds) Air-to-air : 6 x AIM-120C/D medium range air-to-air missile 2 x AIM-9M/X short range air-to-air missile Air-to-ground: 2× 1,000 lb (450 kg) JDAM <i>or</i> 8× 250 lb (110 kg) GBU-39 SDB 2× AIM-120 AMRAAM 2× AIM-9 Sidewinder Hardpoint(external) 4× under-wing pylon stations can be fitted to carry weapons, each with a capacity of 5,000 lb (2,270 kg) <i>or</i> 600 U.S. gallon (2,270 L) drop tanks^[299] 4x AIM-120 AMRAAM (external)	54
F-35 Lightning II	Yes	Guns: 1 x 25,, rotary cannon (180 rounds) Air-to-air: AIM-9X Sidewinder	6 active 48 planned for the next decade

		AIM-120 AMRAAM AIM-132 ASRAAM AIM-260 JATM (to be integrated) Air-to-surface: AGM-88G AARGM-ER AGM-158 JASSM AGM-179 JAGM SPEAR 3 Stand-in Attack Weapon Anti-ship missiles: Agm-158C LRASM Joint Strike Missile Bombs: Joint Direct Attack Munition Paveway Precision-guided glide bomb B61 mod 12 nuclear bomb	
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